

The Endocrine System

- Endocrine and nervous systems work together
- Endocrine system
 - hormones released into the bloodstream travel throughout the body
 - results may take hours, but last longer
- Nervous system
 - certain parts release hormones into blood
 - rest releases neurotransmitters excite or inhibit nerve, muscle & gland cells
 - results in milliseconds, brief duration of effects

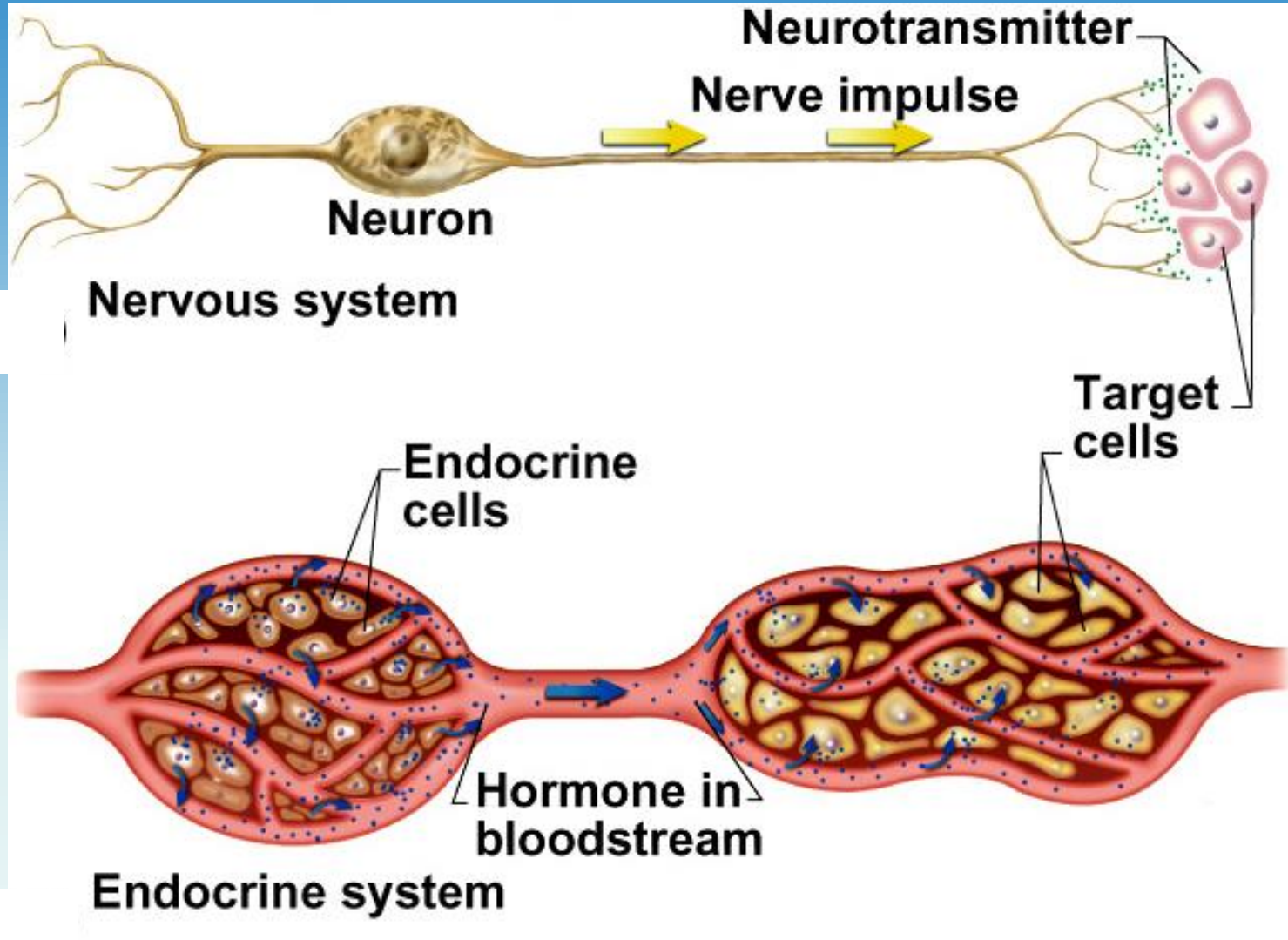
Overview of Cell Communications

- Necessary for integration of cell activities
- Mechanisms
 - gap junctions
 - pores in cell membrane along which signalling chemicals move from cell to cell
 - neurotransmitters
 - released from neurons to travel across gap to 2nd cell
 - paracrine (local) hormones
 - secreted into tissue fluids to effect nearby cells
 - hormones (strict definition)
 - chemical messengers that travel in the bloodstream

Differences in Nervous and Endocrine Systems

- Means of communication
 - nervous system has both electrical and chemical methods
 - endocrine system has only chemical methods
- Speed and persistence of response
 - nervous system reacts quickly (1 - 10 msec) and stops quickly
 - endocrine system reacts slowly (hormone release in seconds or days), effect may continue for weeks
- Adaptation to long-term stimuli
 - nervous system adapts quickly and response declines
 - endocrine system has more persistent responses
- Area of effect
 - nervous system effects are targeted and specific (one organ)
 - endocrine system may have general, widespread effects on many organs

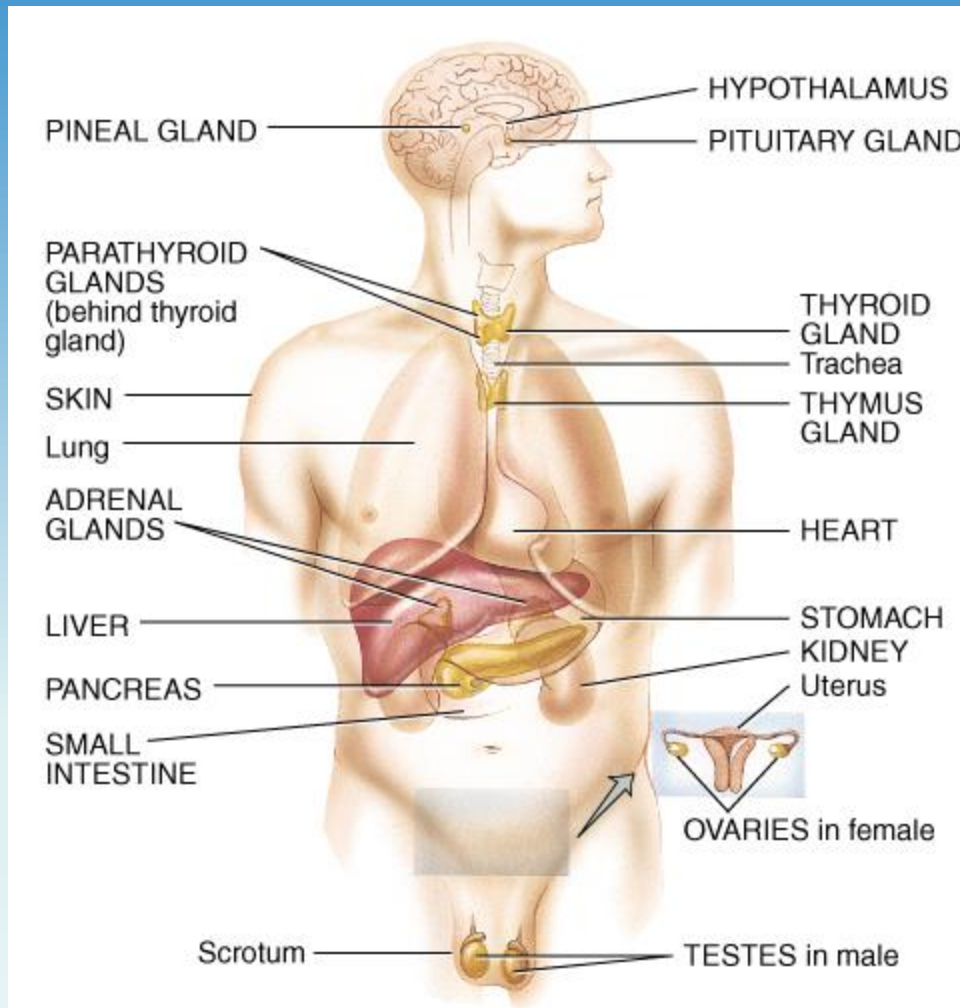
Communication by the Nervous & Endocrine Systems



Similarities in Nervous and Endocrine Systems

- Several chemicals function as both hormones and neurotransmitters
 - norepinephrine, cholecystikinin, thyrotropin-releasing hormone, dopamine and antidiuretic hormone
- Some hormones secreted by neuroendocrine cells (neurons)
 - oxytocin and catecholamines
- Both systems with overlapping effects on same target cells
 - norepinephrine and glucagon cause glycogen hydrolysis in liver
- Systems regulate each other
 - neurons trigger hormone secretion
 - hormones stimulate or inhibit neurons

General Functions of Hormones



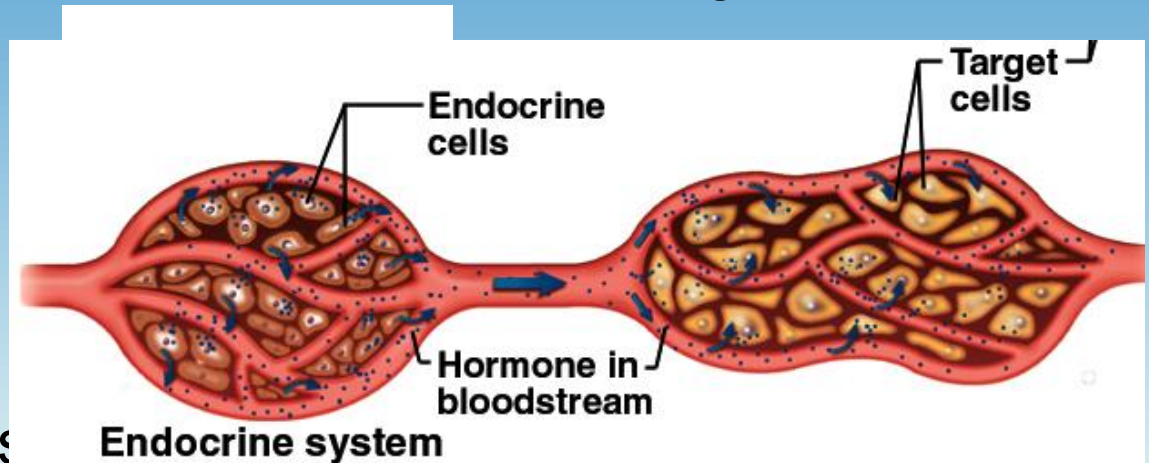
- Help regulate:
 - extracellular fluid
 - metabolism
 - biological clock
 - contraction of cardiac & smooth muscle
 - glandular secretion
 - some immune functions
- Growth & development
- Reproduction

Endocrine Glands Defined

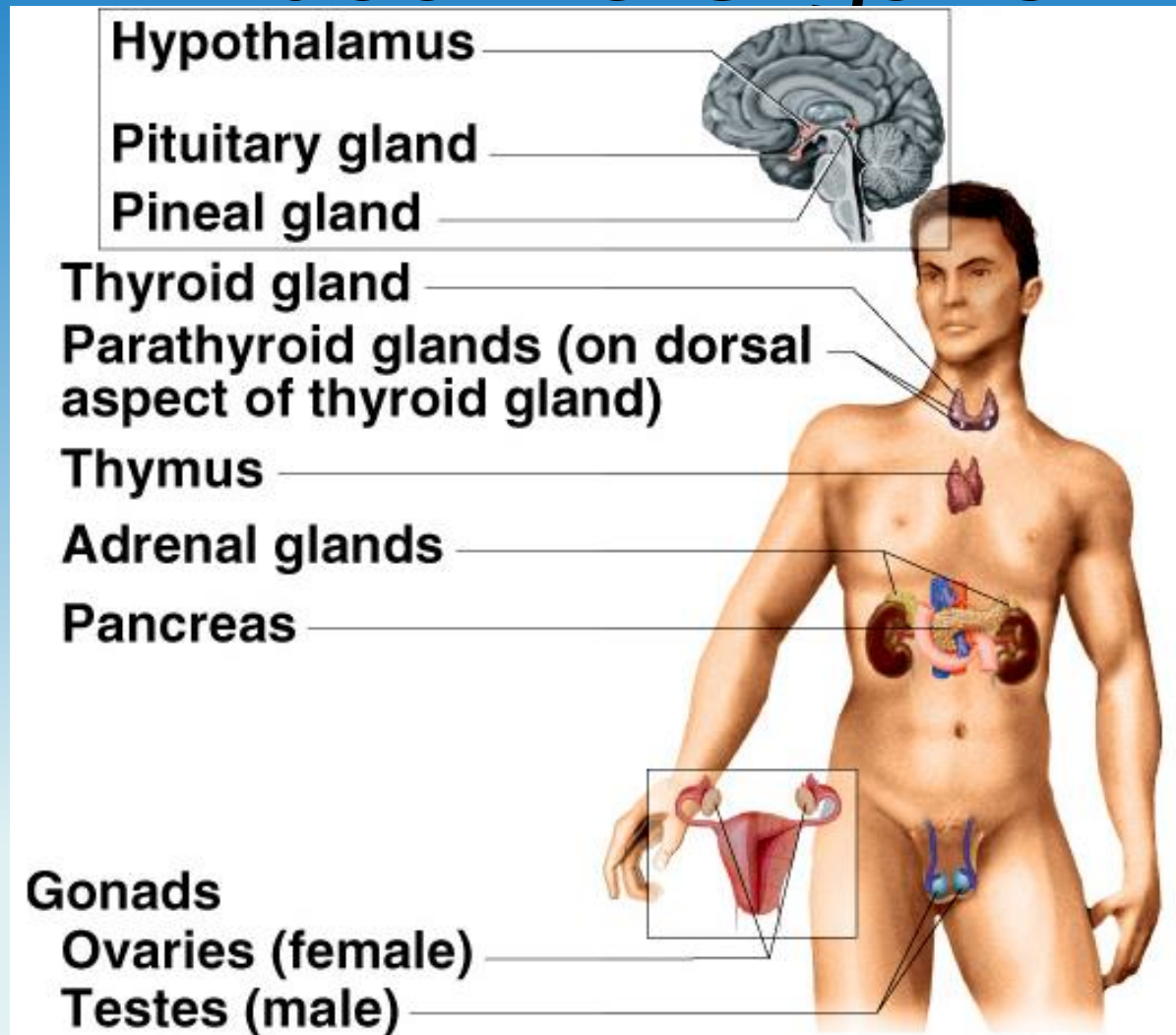
- Exocrine glands
 - secrete products into ducts which empty into body cavities or body surface
 - Extracellular effects
 - sweat, oil, mucous, & digestive glands
- Endocrine glands
 - secrete products (hormones) into bloodstream
 - Intracellular effects
 - pituitary, thyroid, parathyroid, adrenal, pineal
 - other organs secrete hormones as a 2nd function
 - hypothalamus, thymus, pancreas, ovaries, testes, kidneys, stomach, liver, small intestine, skin, heart & placenta

Components of Endocrine System

- Hormone
 - chemical messenger secreted into bloodstream, stimulates response in another tissue or organ
- Target cells
 - have receptors for hormone
- Endocrine glands
 - produce hormones
- Endocrine system
 - includes endocrine organs (thyroid, pineal, etc)
 - includes hormone producing cells in organs such as brain, heart and small intestine



Endocrine Organs

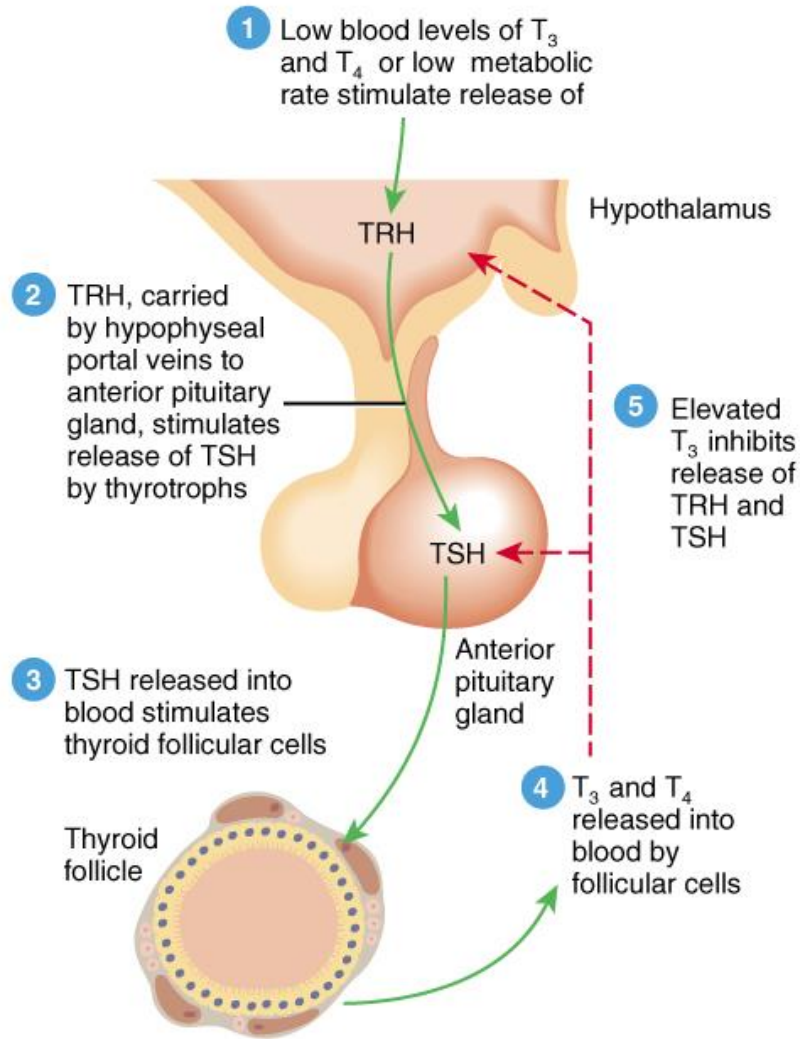


- Major organs of endocrine system

Control of Hormone Secretion

- Regulated by signals from nervous system, chemical changes in the blood or by other hormones
- Negative feedback control (most common)
 - decrease/increase in blood level is reversed
- Positive feedback control
 - the change produced by the hormone causes more hormone to be released
- Disorders involve either hyposecretion or hypersecretion of a hormone
- hormone release in response to higher brain centers
 - milk ejection reflex can be triggered by a baby's cry

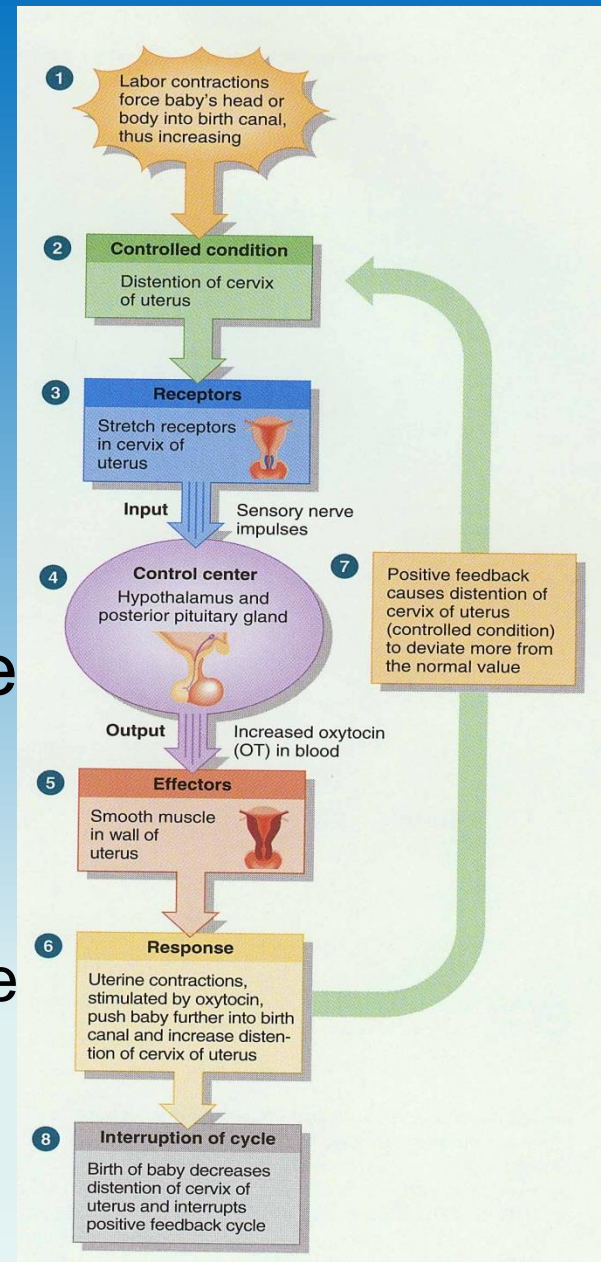
Negative Feedback Systems



- Decrease in blood levels
- Receptors in hypothalamus & thyroid
- Cells activated to secrete more TSH or more T_3 & T_4
- Blood levels increase

Positive Feedback

- Oxytocin stimulates uterine contractions
- Uterine contractions stimulate oxytocin release
- hormone release in response to nervous system signals
 - suckling infant → stimulates nerve endings → hypothalamus → posterior lobe → oxytocin → milk ejection



Hypothalamus and Pituitary Gland

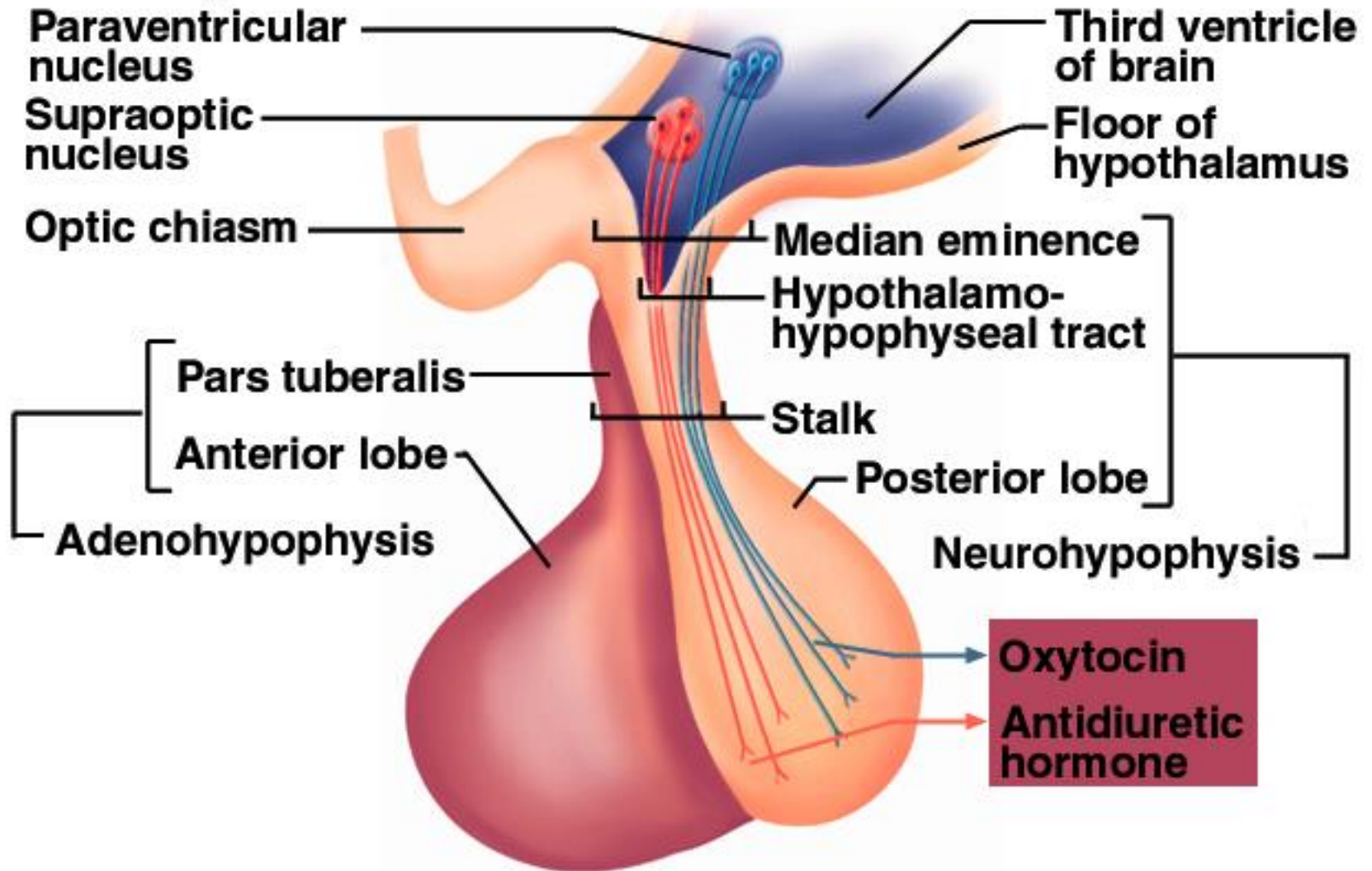
- Both are master endocrine glands since their hormones control other endocrine glands
- Hypothalamus is a section of brain above where pituitary gland is suspended from stalk
- Hypothalamus receives input from cortex, thalamus, limbic system & internal organs
- Hypothalamus controls pituitary gland with 9 different releasing & inhibiting hormones

Anatomy of Pituitary Gland

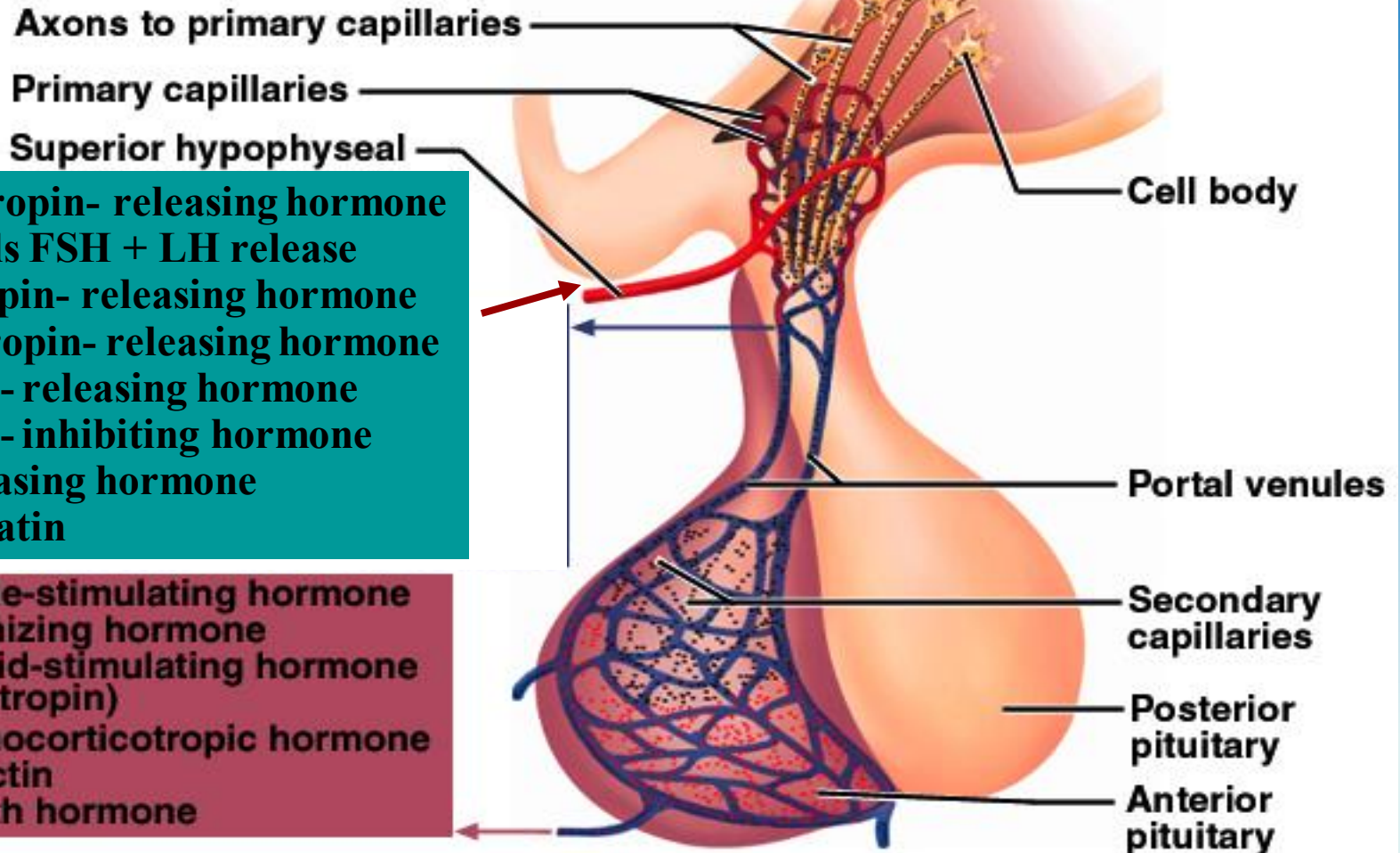
- Pea-shaped, 1/2 inch gland found in sella turcica of sphenoid
- Infundibulum attaches it to brain
- Anterior lobe = 75%
develops from roof of mouth
- Posterior lobe = 25%
 - ends of axons of 10,000 neurons found in hypothalamus
 - neuroglial cells called pituicytes

Pituitary Gland Anatomy and Hormones of the Neurohypophysis

Nuclei of hypothalamus



Hypothalamo-Hypophyseal Portal System

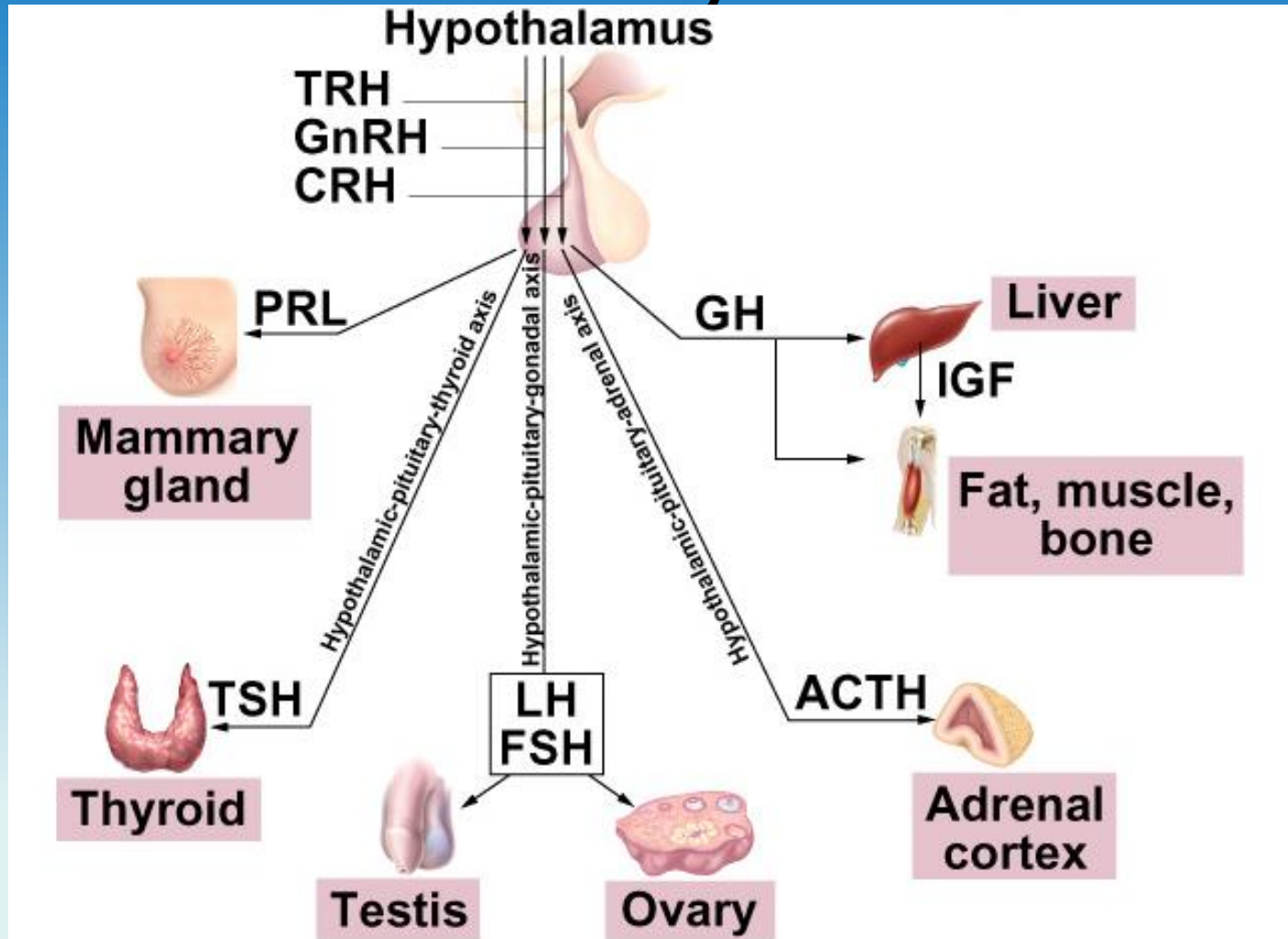


- Hormones secreted by hypothalamus, travel in portal system to anterior pituitary
- Hormones (red box) secreted by anterior pituitary (under control of hypothalamic releasers and inhibitors)

Pituitary Hormones - Anterior Lobe

- Tropic hormones target other endocrine glands
 - gonadotropins target gonads, FSH (follicle stimulating hormone) and LH (luteinizing hormone)
 - TSH (thyroid stimulating hormone)
 - ACTH (adrenocorticotrophic hormone)
- PRL (prolactin)
- GH (growth hormone)

Anterior Pituitary Hormones

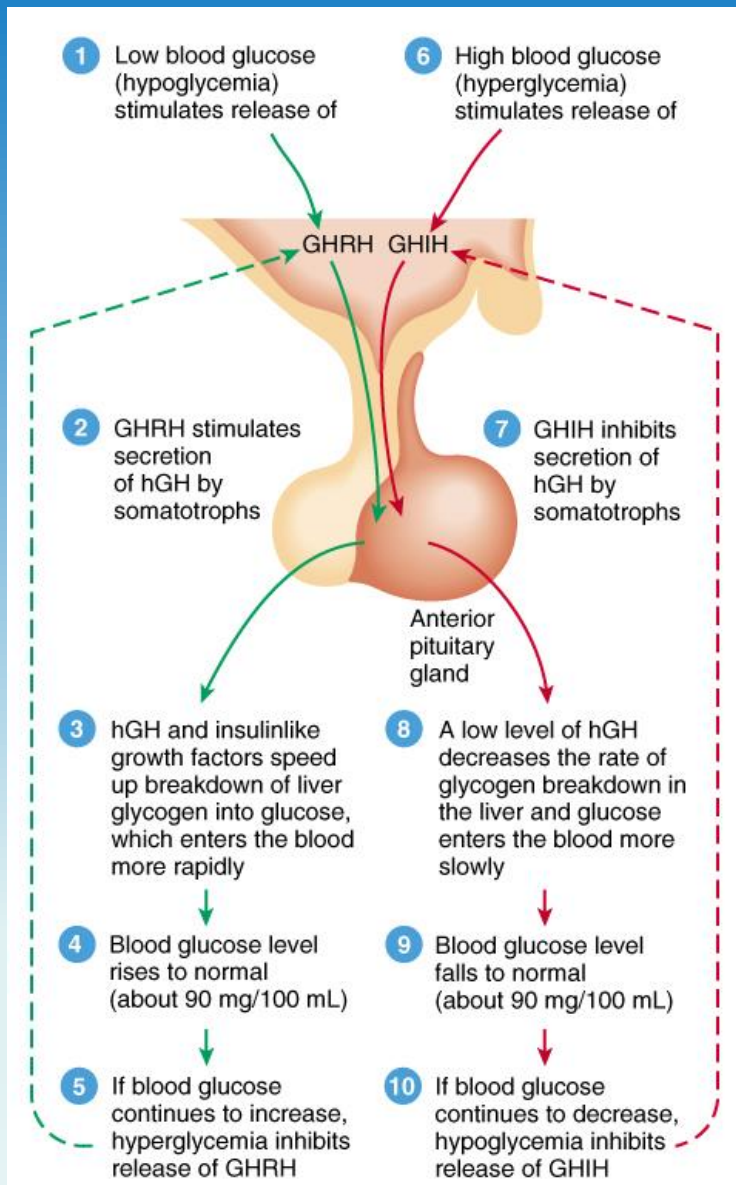


- Principle hormones and target organs shown
- Axis - refers to way endocrine glands interact

Growth Hormone

- Secreted by somatotropes of anterior pituitary
- Promotes tissue growth
 - directly affects mitosis and cellular differentiation
 - indirectly stimulates liver to produce IGF-I somatomedins
 - half-life of 20 hours compared to 20 minutes for GH
- Functions of GH-IGF
 - protein synthesis
 - $\uparrow$ DNA transcription for $\uparrow$ mRNA production, proteins synthesized
 - enhances amino acid transport into cells, $\downarrow$ protein catabolism
 - lipid metabolism
 - stimulates FFA and glycerol release from adipocytes, protein sparing
 - CHO metabolism
 - glucose sparing effect = less glucose used for energy
 - Electrolyte balance
 - promotes Na^+ , K^+ , Cl^- retention, Ca^{+2} absorption

Regulation of hGH



- Low blood sugar stimulates release of GHRH from hypothalamus
 - anterior pituitary releases more hGH, more glycogen broken down into glucose by liver cells
- High blood sugar stimulates release of GHIH from hypothalamus
 - less hGH from anterior pituitary, glycogen does not breakdown into glucose

Diabetogenic Effect of Human Growth Hormone

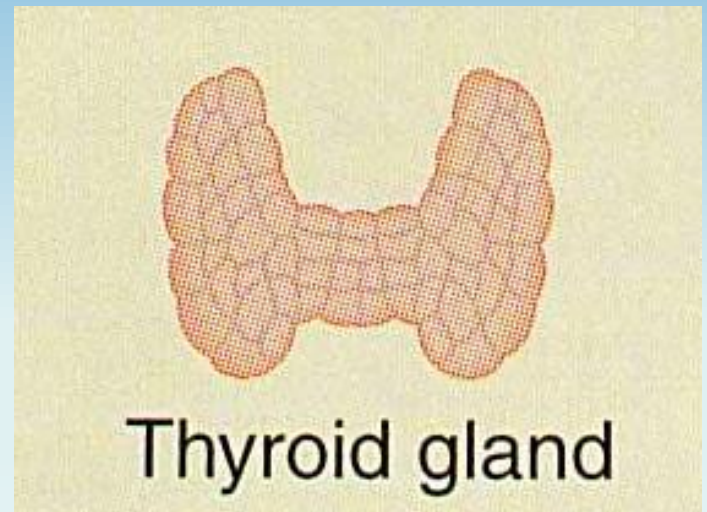
- Excess of growth hormone
 - raises blood glucose concentration
 - pancreas releases insulin continually
 - beta-cell burnout
- Diabetogenic effect
 - causes diabetes mellitus if no insulin activity can occur eventually

Growth Hormone and Aging

- Childhood and adolescence
 - bone, cartilage and muscle growth
- Adulthood
 - increase osteoblastic activity and appositional growth affecting bone thickening and remodeling
 - blood concentration decrease by age 75 to $\frac{1}{4}$ of that of adolescent
- Levels of GH
 - higher during first 2 hours of deep sleep, after high protein meals, after vigorous exercise
 - lower after high CHO meals

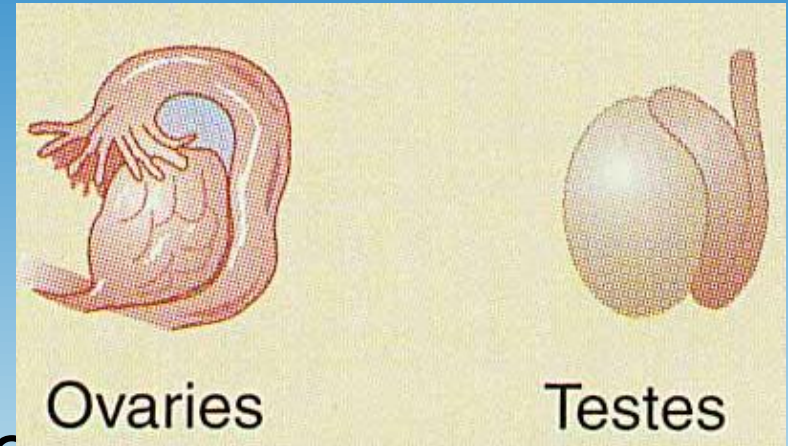
Thyroid Stimulating Hormone (TSH)

- Hypothalamus regulates thyrotroph cells
- Thyrotroph cells produce TSH
- TSH stimulates the synthesis & secretion of T3 and T4
- Metabolic rate stimulated



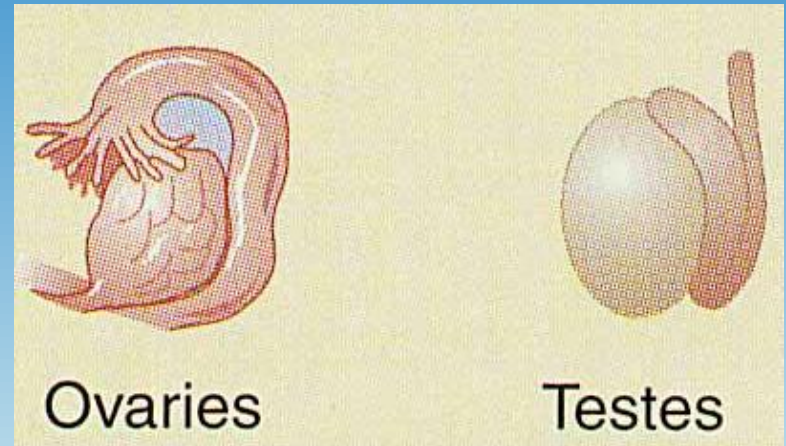
Follicle Stimulating Hormone (FSH)

- Releasing hormone from hypothalamus controls gonadotrophs
- Gonadotrophs release follicle stimulating hormone
- FSH functions
 - initiates the formation of follicles within the ovary
 - stimulates follicle cells to secrete estrogen
 - stimulates sperm production in testes



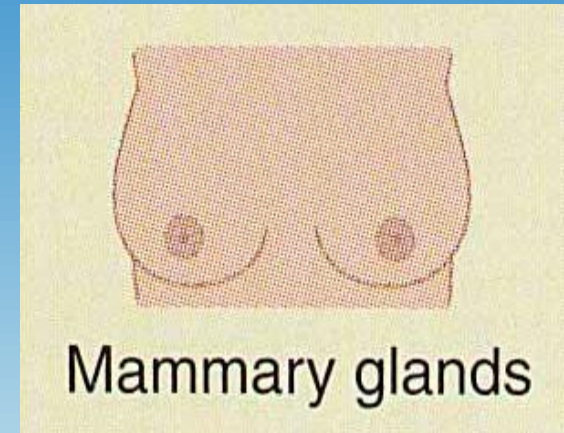
Luteinizing Hormone (LH)

- Releasing hormones from hypothalamus stimulate gonadotrophs
- Gonadotrophs produce LH
- In females, LH stimulates
 - secretion of estrogen
 - ovulation of oocyte from ovary
 - formation of corpus luteum
 - secretion of progesterone
- In males, stimulates interstitial cells to secrete testosterone

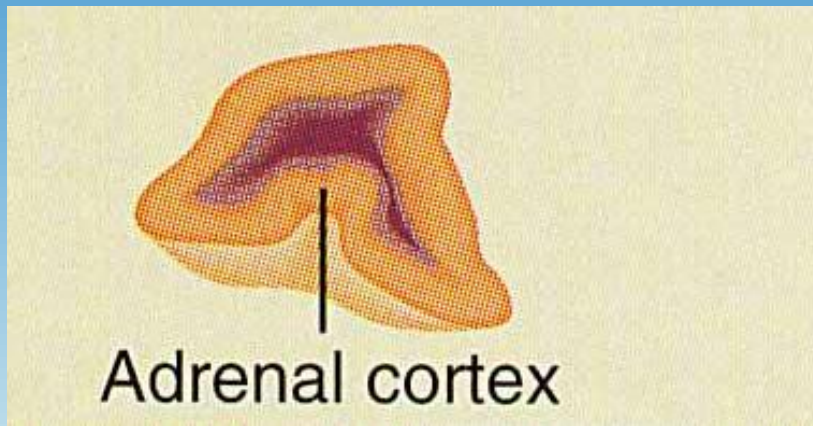


Prolactin (PRL)

- Hypothalamus regulates lactotroph cells
- Lactotrophs produce prolactin
- Under right conditions, prolactin causes milk production
- Suckling reduces levels of hypothalamic inhibition and prolactin levels rise along with milk production
- Nursing ceases & milk production slows



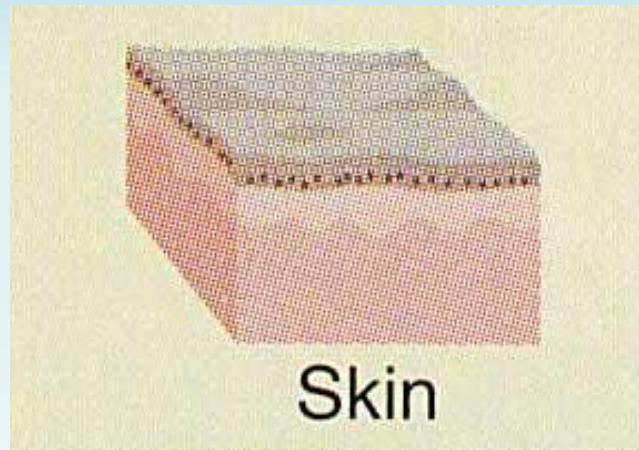
Adrenocorticotrophic Hormone



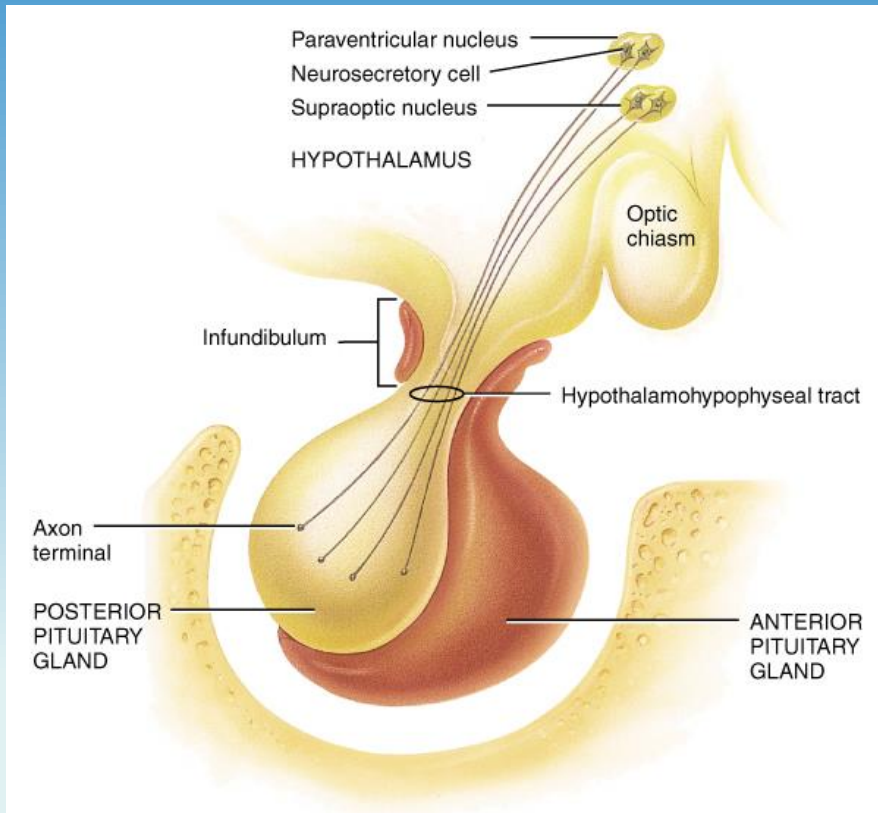
- Regulates response to stress
- Hypothalamus releasing hormones stimulate corticotrophs
- Corticotrophs secrete ACTH & MSH
- ACTH stimulates cells of the adrenal cortex that produce glucocorticoids

Melanocyte-Stimulating Hormone

- Secreted by corticotroph cells
- Releasing hormone from hypothalamus increases its release From the anterior pituitary
- Function not certain in humans (increase skin pigmentation in frogs)



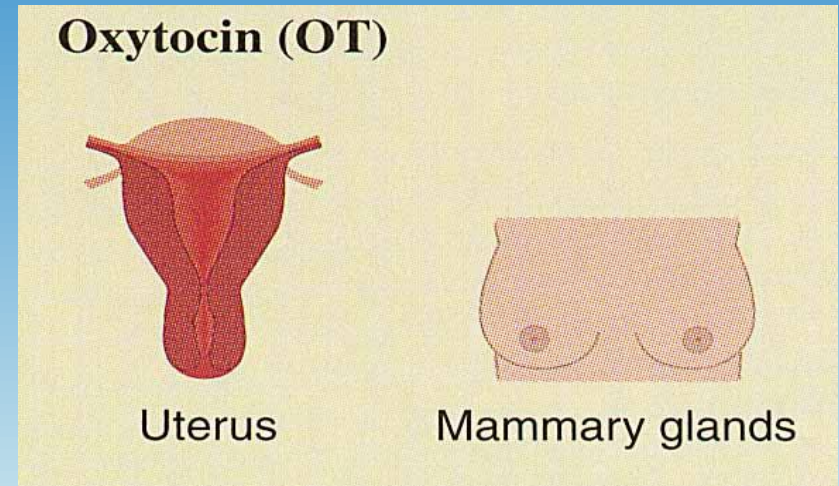
Posterior Pituitary Gland (Neurohypophysis)



- Does not synthesize hormones
- Consists of axon terminals of hypothalamic neurons
- Neurons release two neurotransmitters that enter capillaries
 - antidiuretic hormone
 - oxytocin

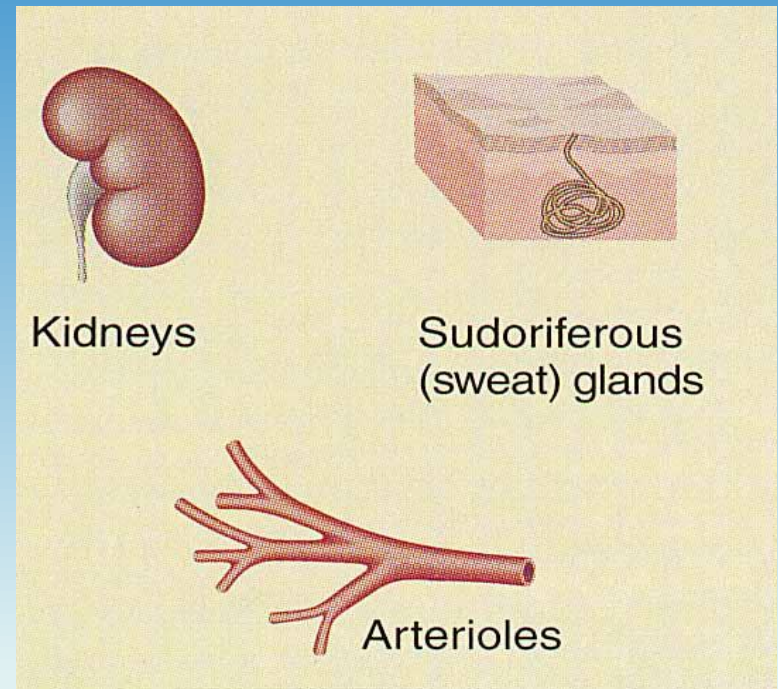
Oxytocin

- Two target tissues both involved in neuroendocrine reflexes
- During delivery
 - baby's head stretches cervix
 - hormone release enhances uterine muscle contraction
 - baby & placenta are delivered
- After delivery
 - suckling & hearing baby's cry stimulates milk ejection
 - hormone causes muscle contraction & milk ejection
- Males ???

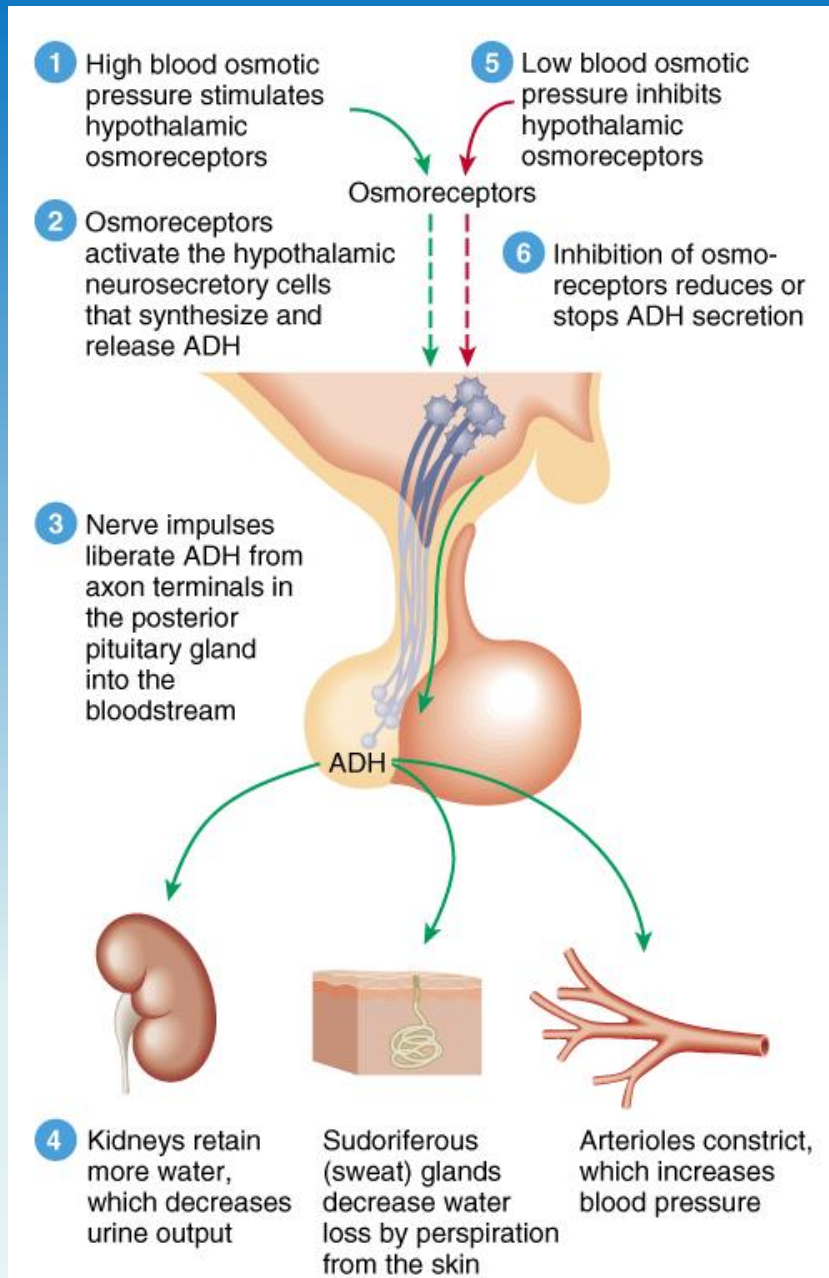


Antidiuretic Hormone (ADH)

- Known as vasopressin
- Functions
 - decrease urine production
 - decrease sweating
 - increase BP



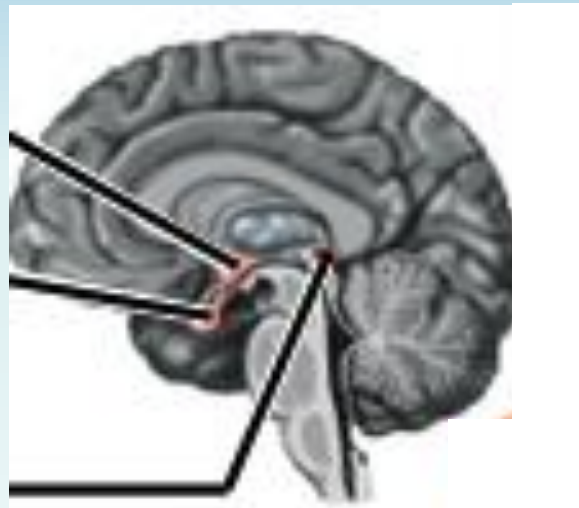
Regulation of ADH



- Dehydration
 - ADH released
- Overhydration
 - ADH inhibited

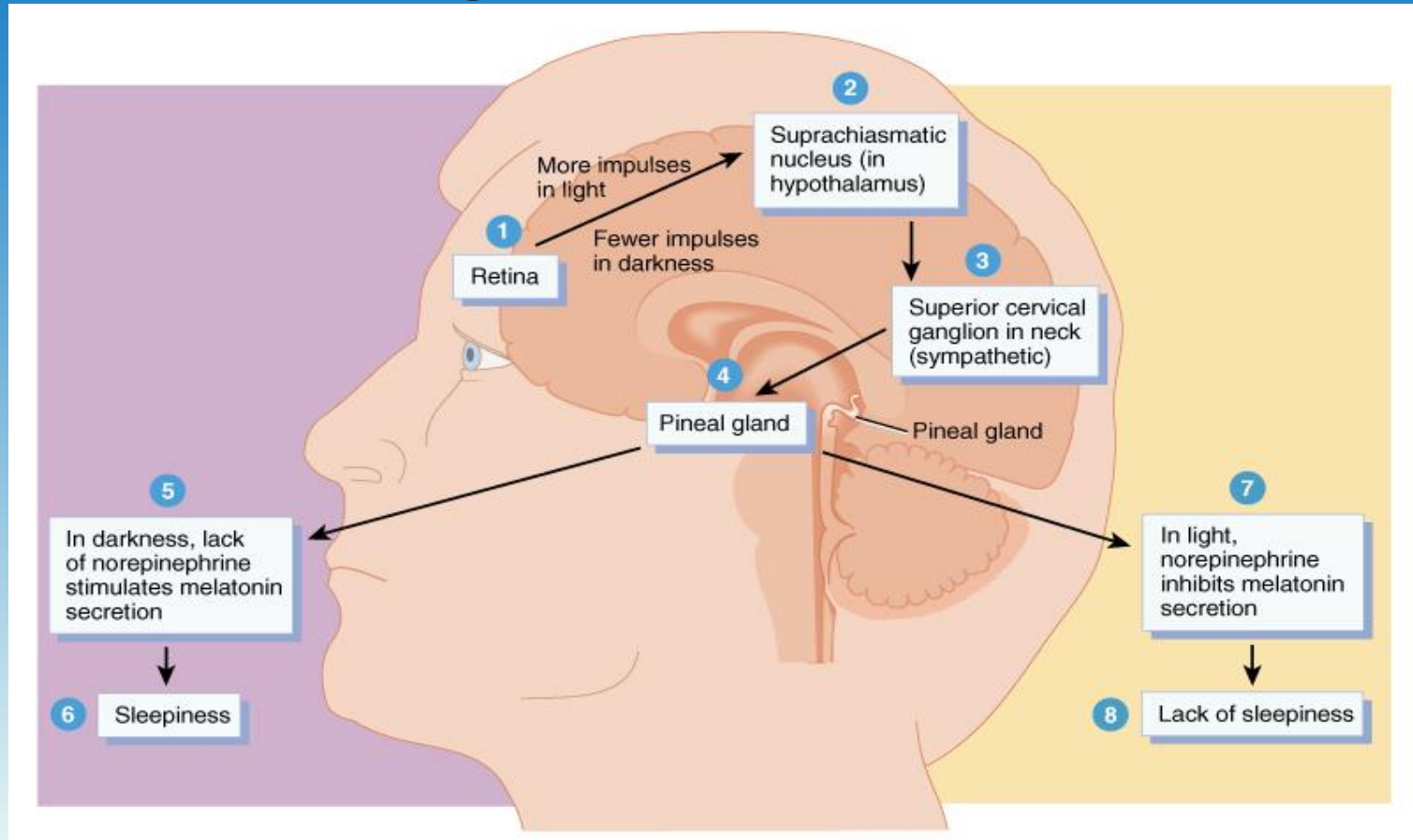
Pineal Gland

- Peak secretion 1-5 yr. olds, by puberty 75% lower
- Produces serotonin by day, converts it to melatonin at night
- May regulate timing of puberty in humans
- Melatonin $\uparrow$ in SAD + PMS, $\downarrow$ by phototherapy
 - depression, sleepiness, irritability and carbohydrate craving



Pineal gland

Effect of Light on Pineal Gland



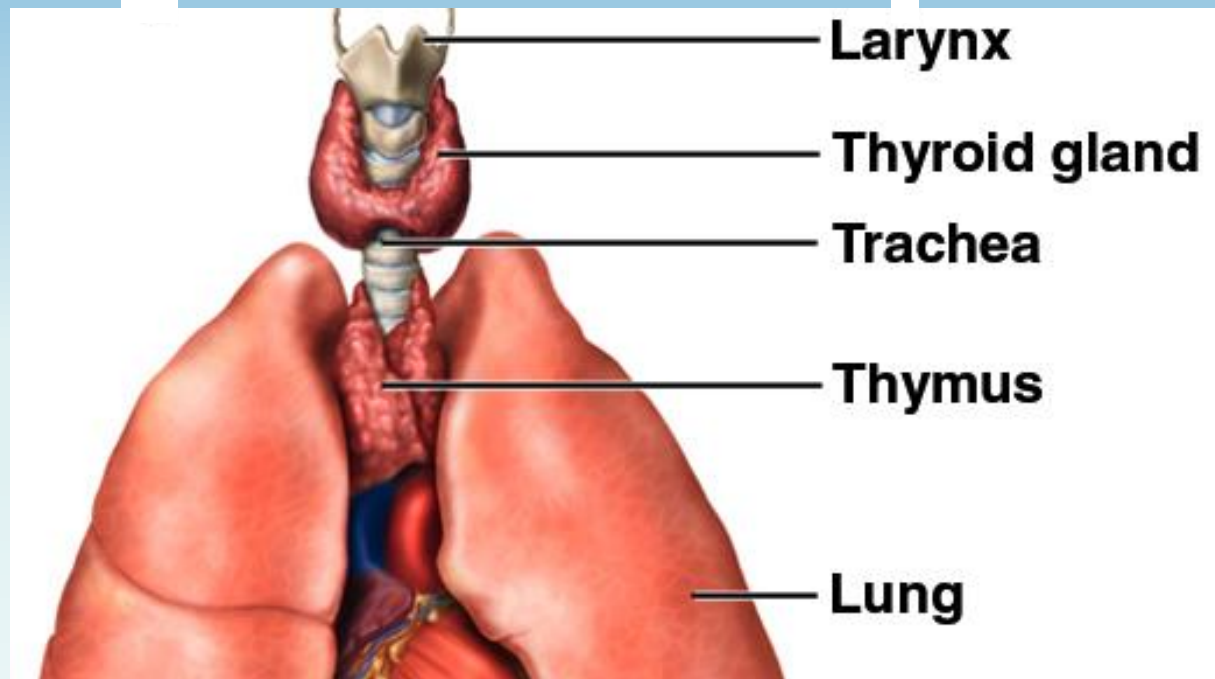
- Melatonin secretion producing sleepiness occurs during darkness due to lack of stimulation from sympathetic ganglion

Seasonal Affective Disorder and Jet Lag

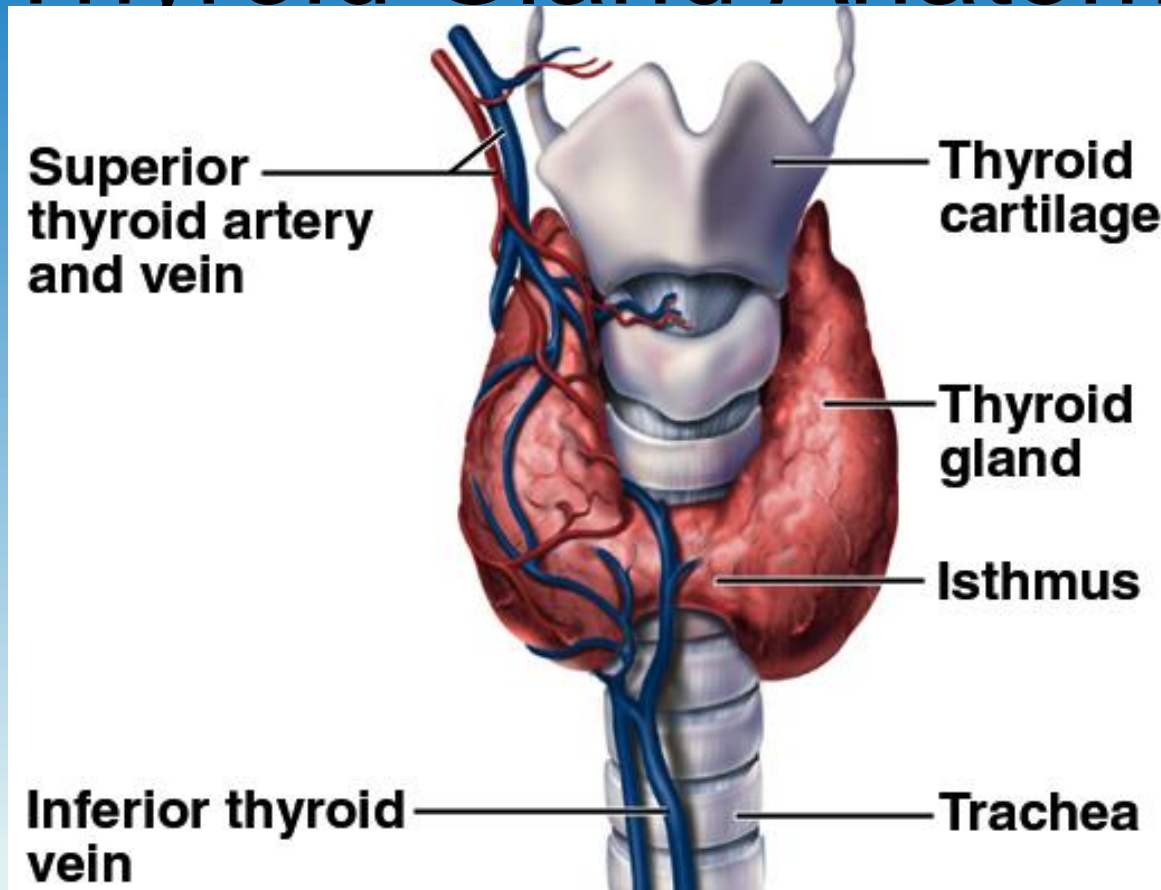
- Depression that occurs during winter months when day length is short
- Due to overproduction of melatonin
- Therapy
 - exposure to several hours per day of artificial light as bright as sunlight
 - speeds recovery from jet lag

Thymus

- Location: mediastinum, superior to heart
- Involution after puberty
- Secretes hormones that regulate development and later activation of T-lymphocytes
 - thymopoietin and thymosins



Thyroid Gland Anatomy

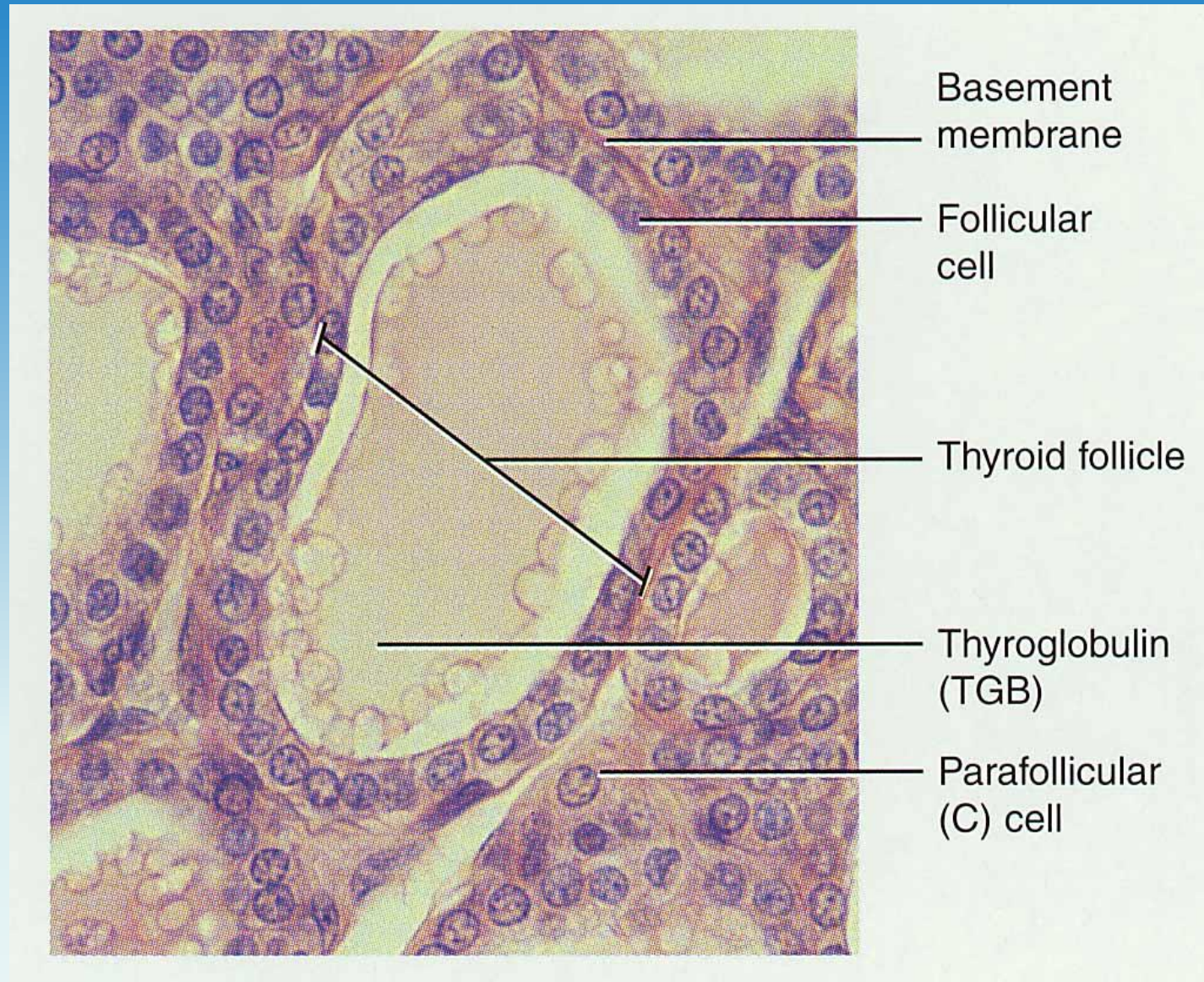


- Largest endocrine gland with high rate of blood flow
- Anterior and lateral sides of trachea
- 2 large lobes connected by isthmus

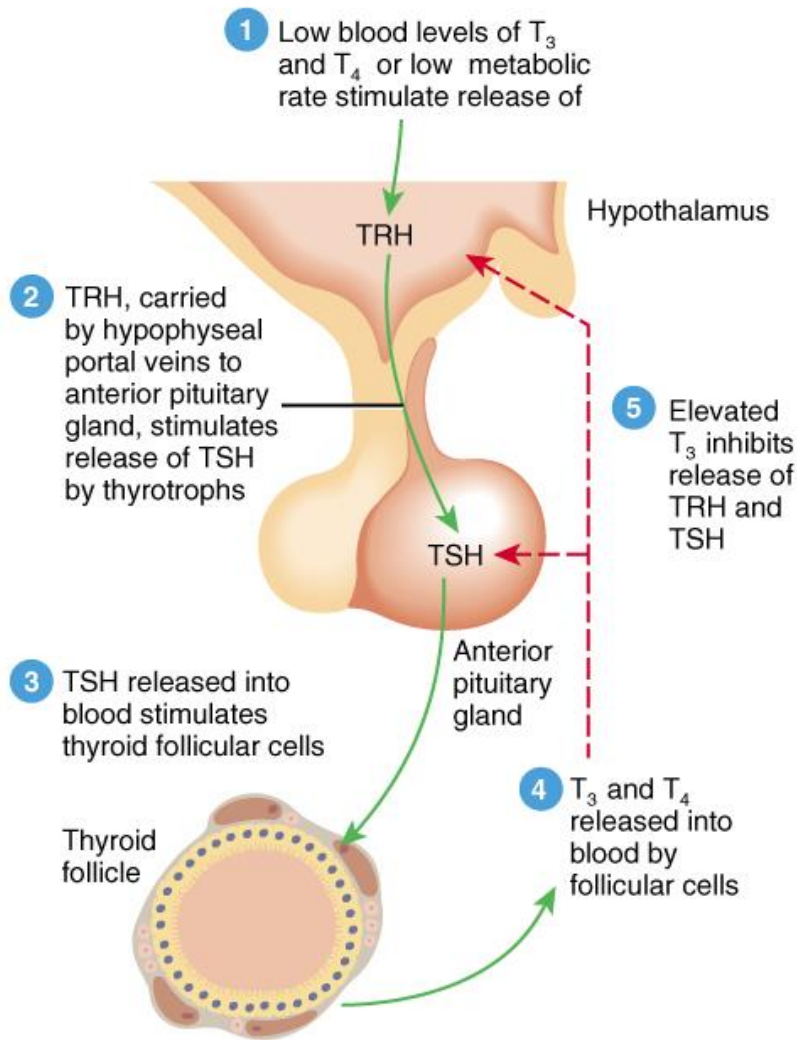
Thyroid Gland

- Thyroid follicles
 - filled with colloid and lined with simple cuboidal epith. (follicular cells) that secretes 2 hormones, $T_3 + T_4$
 - Thyroid hormone
 - $\uparrow$ body's metabolic rate and O_2 consumption
 - calorogenic effect - $\uparrow$ heat production
 - $\uparrow$ heart rate and contraction strength
 - $\uparrow$ respiratory rate
 - stimulates appetite and breakdown CHO, lipids & proteins
- C (calcitonin or parafollicular) cells
 - produce calcitonin that $\downarrow$ blood Ca^{+2} , promotes Ca^{+2} deposition and bone formation especially in children

Photomicrograph of Thyroid Gland



Control of T3 & T4 Secretion

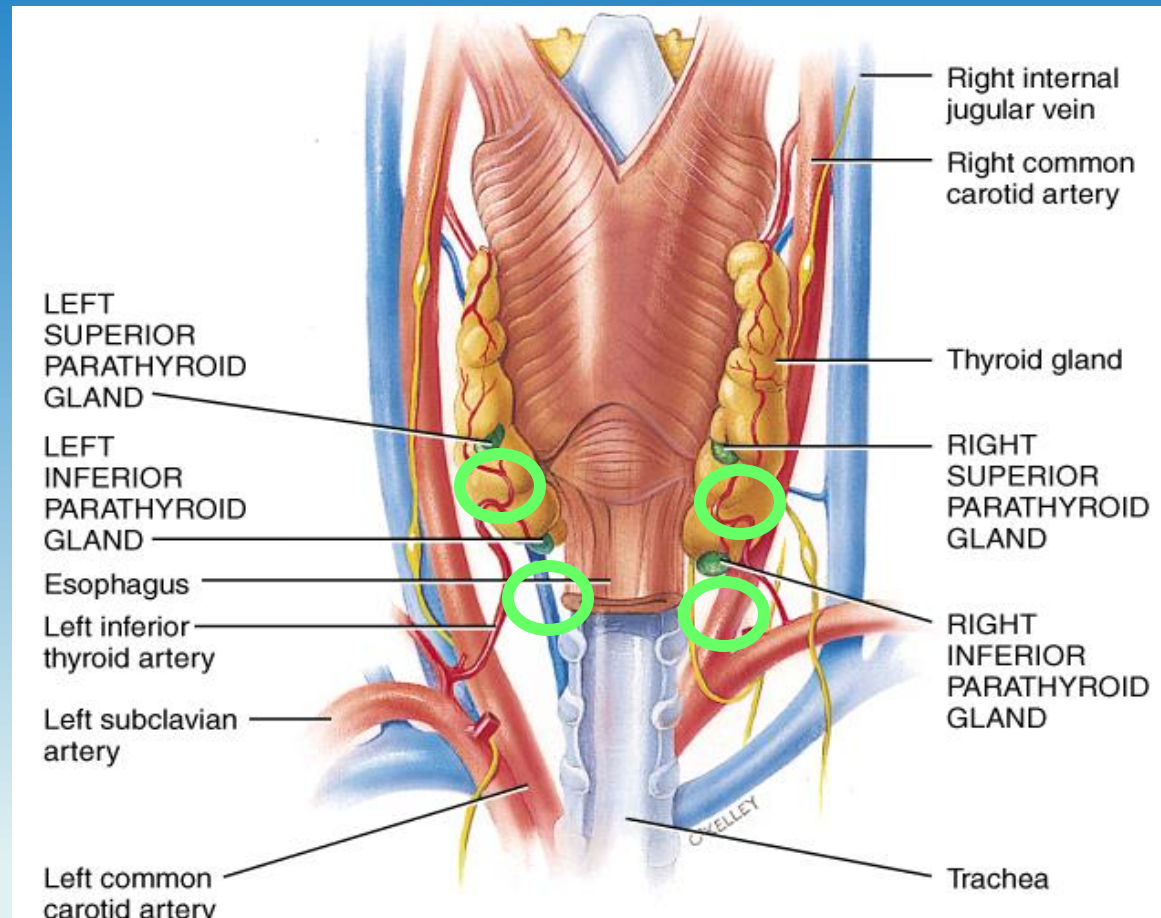


Key:

TRH = Thyrotropin releasing hormone
TSH = Thyroid-stimulating hormone
 T_3 = Triiodothyronine
 T_4 = Thyroxine (Tetraiodothyronine)

- Negative feedback system
- Low blood levels of hormones stimulate hypothalamus
- It stimulates pituitary to release TSH
- TSH stimulates gland to raise blood levels

Parathyroid Glands

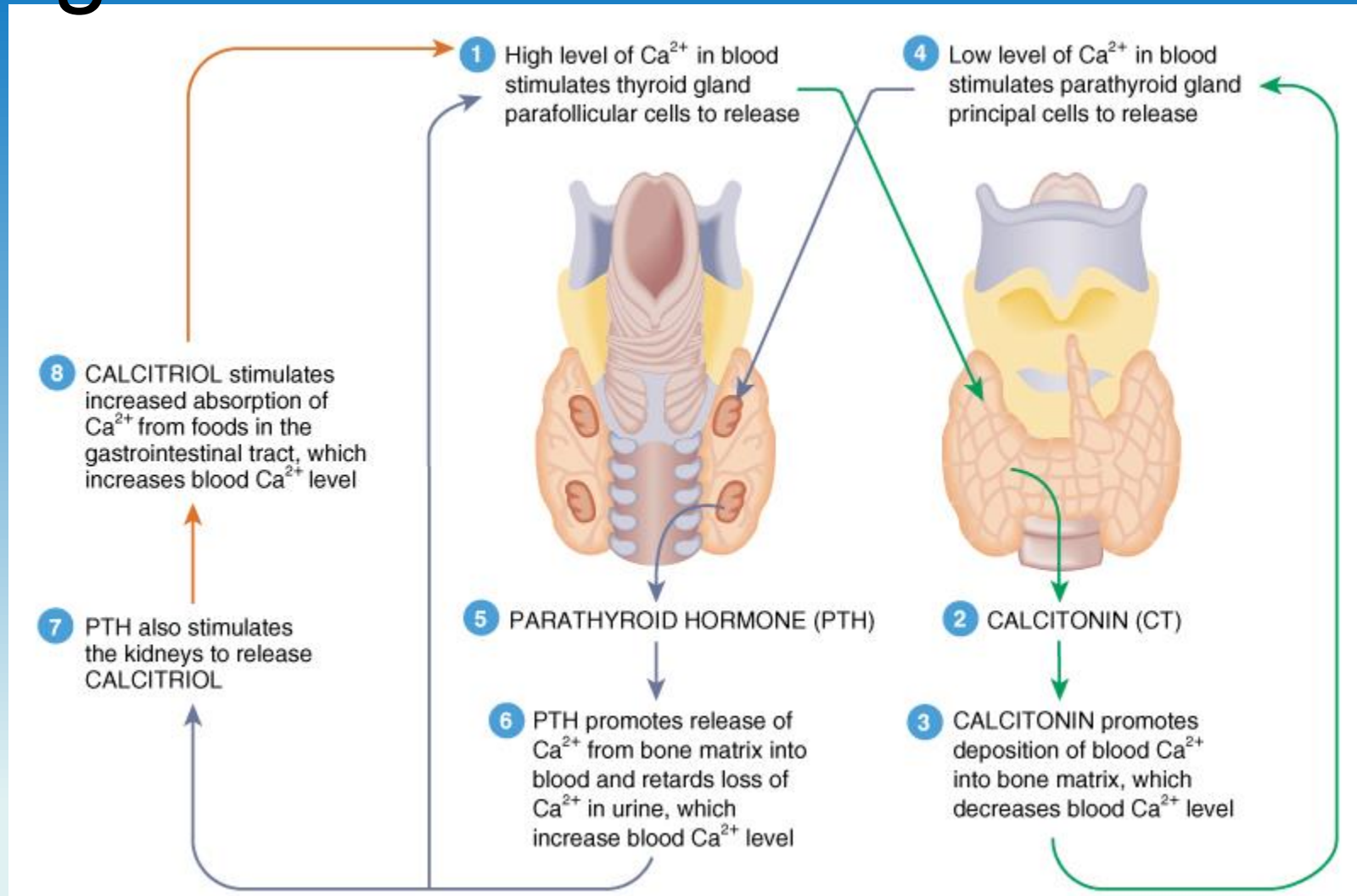


- 4 pea-sized glands found on back of thyroid gland

Parathyroid Hormone

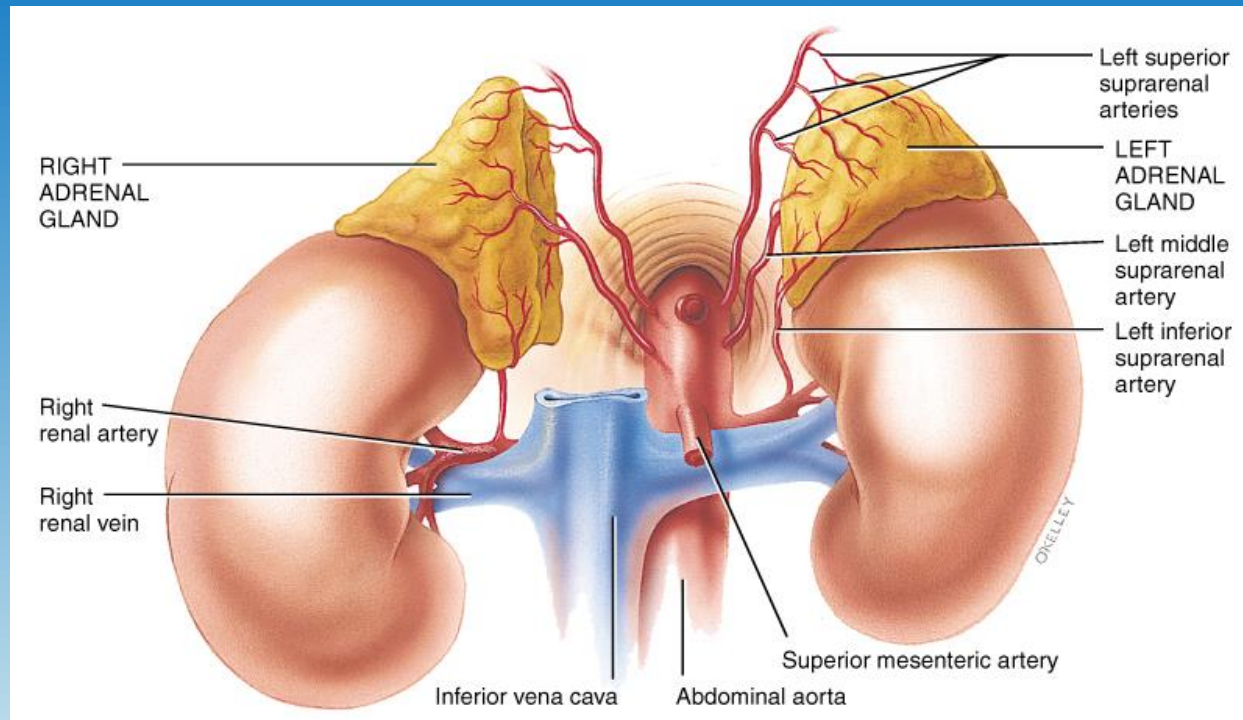
- Raise blood calcium levels
 - increase activity of osteoclasts
 - increases reabsorption of Ca^{+2} by kidney
 - inhibits reabsorption of phosphate (HPO_4) $^{-2}$
 - promote formation of calcitriol (vitamin D3) by kidney which increases absorption of Ca^{+2} and Mg^{+2} by intestinal tract
- Opposite function of calcitonin

Regulation of Calcium Blood Levels



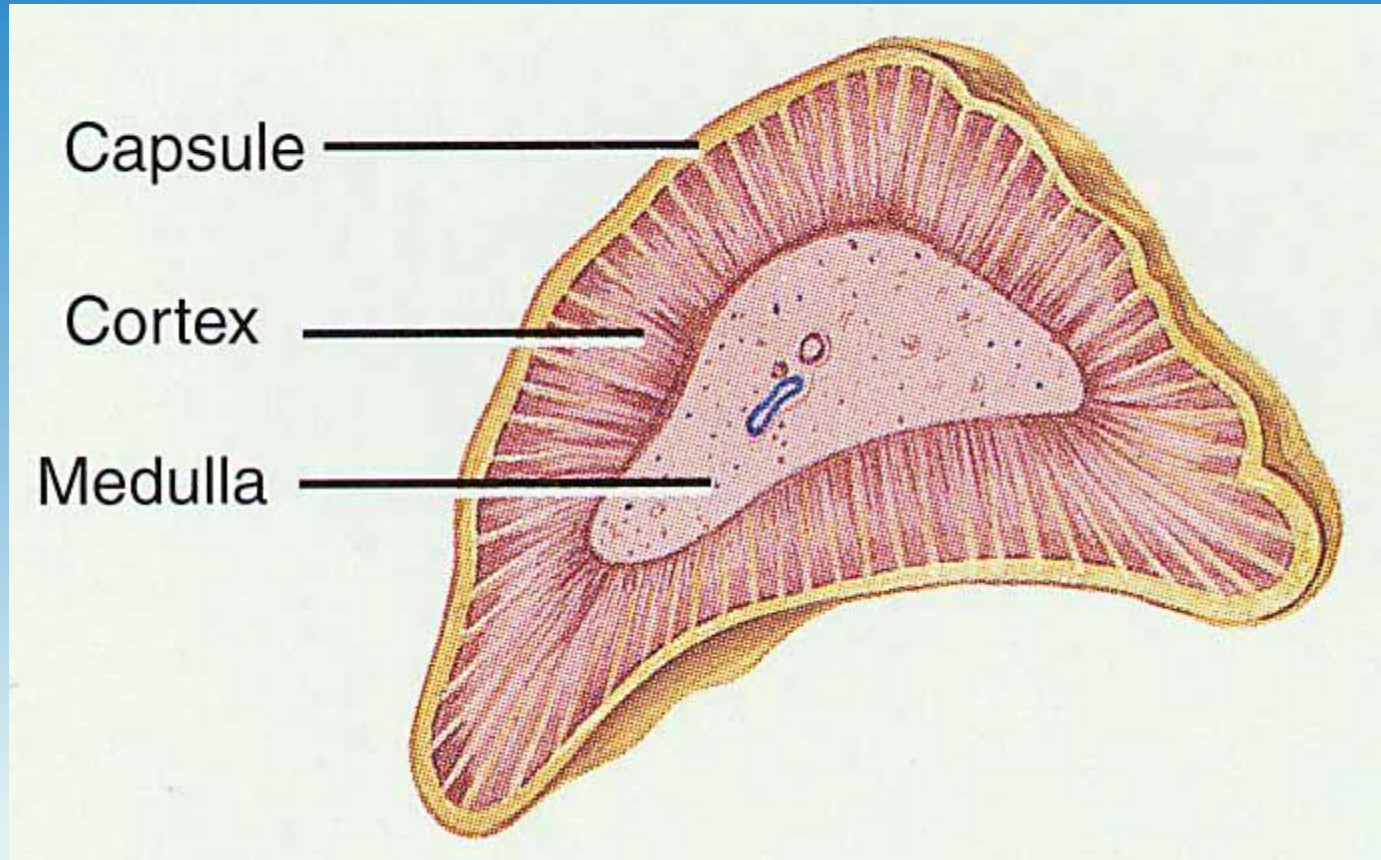
- High or low blood levels of Ca^{2+} stimulate the release of different hormones --- PTH or CT

Adrenal Glands



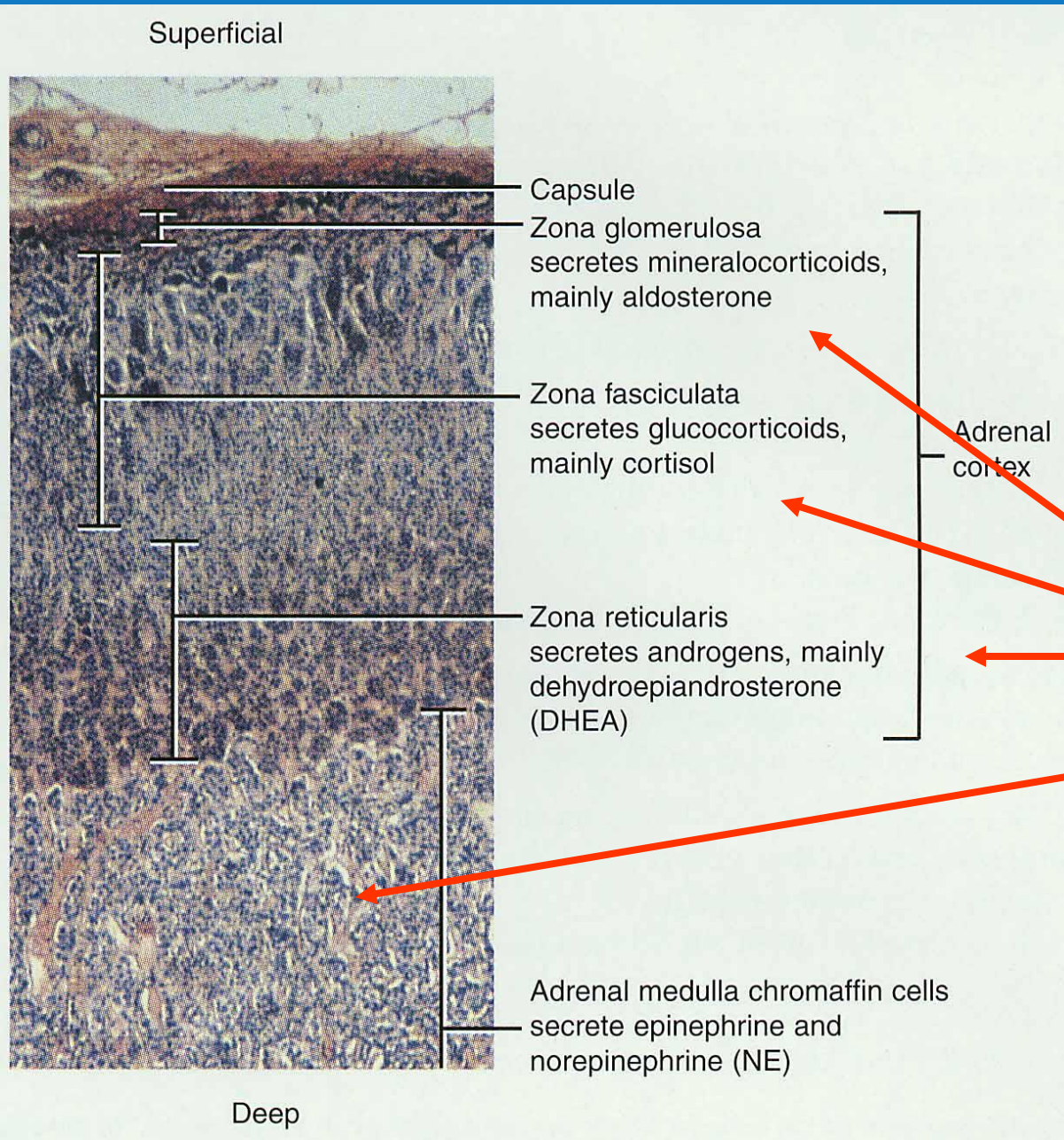
- One on top of each kidney
- 3 x 3 x 1 cm in size and weighs 5 grams
- Cortex produces 3 different types of hormones from 3 zones of cortex
- Medulla produces epinephrine & norepinephrine

Structure of Adrenal Gland



- Cortex derived from mesoderm
- Medulla derived from ectoderm

Histology of Adrenal Gland



- Cortex
 - 3 zones
- Medulla

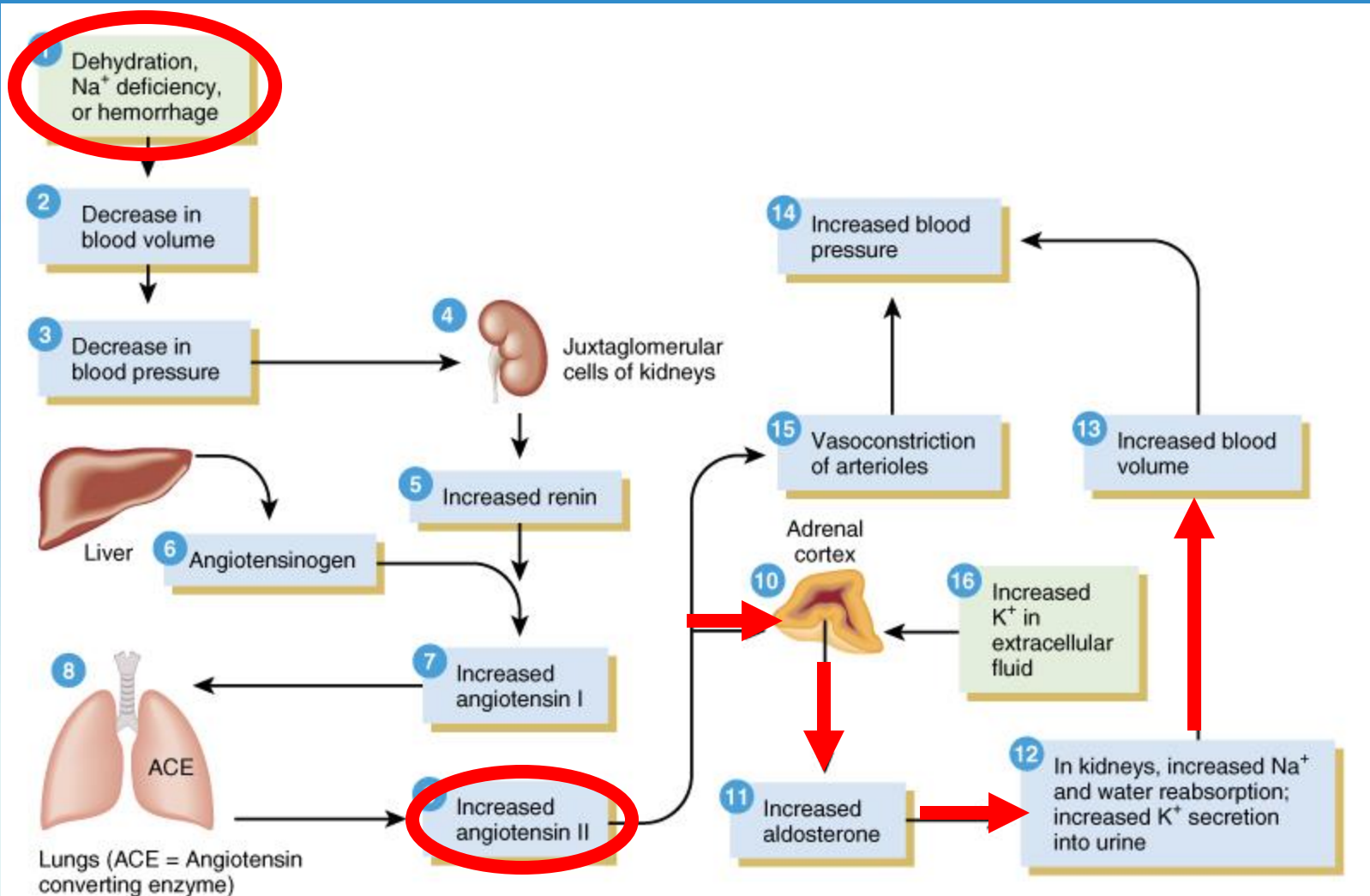
Adrenal Medulla

- Sympathetic ganglion innervated by sympathetic preganglionic fibers
 - consists of modified neurons called chromaffin cells
 - stimulation causes release of (nor-)epinephrine
- Hormonal effect is longer lasting
 - increases BP and heart rate
 - increases blood flow to skeletal muscle
 - increases pulmonary air flow
 - decreases digestion and urine formation
 - stimulates gluconeogenesis and glycogenolysis
- Stress causes medullary cells to stimulate cortex

Mineralocorticoids

- 95% of hormonal activity due to aldosterone
- Functions
 - increase reabsorption of Na^+ with Cl^- , bicarbonate and water following it
 - promotes excretion of K^+ and H^+
- Hypersecretion = tumor producing aldosteronism
 - high blood pressure caused by retention of Na^+ and water in blood

Regulation of Aldosterone



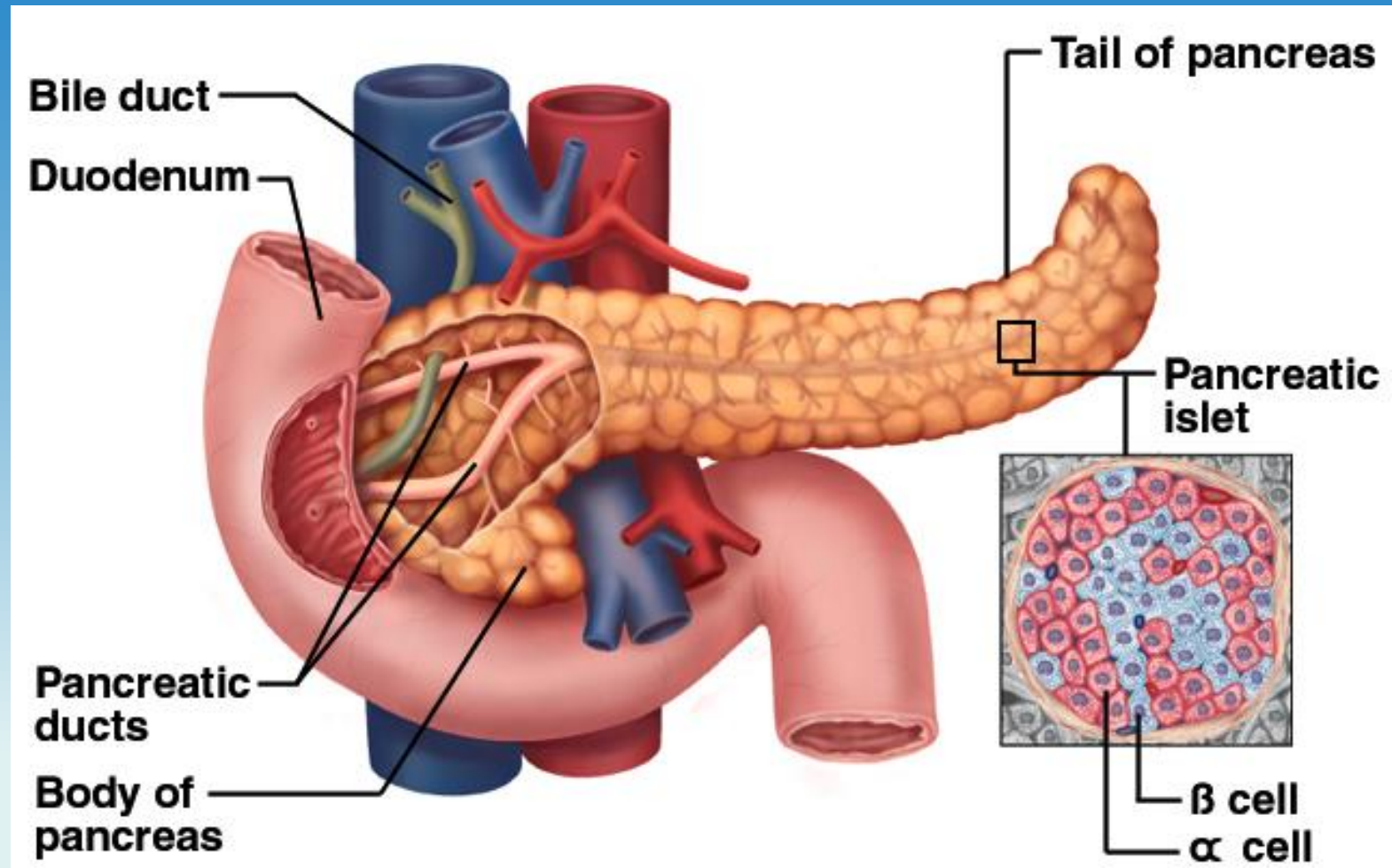
Glucocorticoids

- 95% of hormonal activity is due to cortisol
- Functions = help regulate metabolism
 - increase rate of protein catabolism & lipolysis
 - conversion of amino acids to glucose
 - stimulate lipolysis
 - provide resistance to stress by making nutrients available for ATP production
 - raise BP by vasoconstriction
 - inflammatory effects reduced (skin cream)
 - reduce release of histamine from mast cells
 - decrease capillary permeability
 - depress phagocytosis

Androgens from Zona Reticularis

- Small amount of male hormone produced
 - insignificant in males
 - may contribute to sex drive in females
 - is converted to estrogen in postmenopausal females

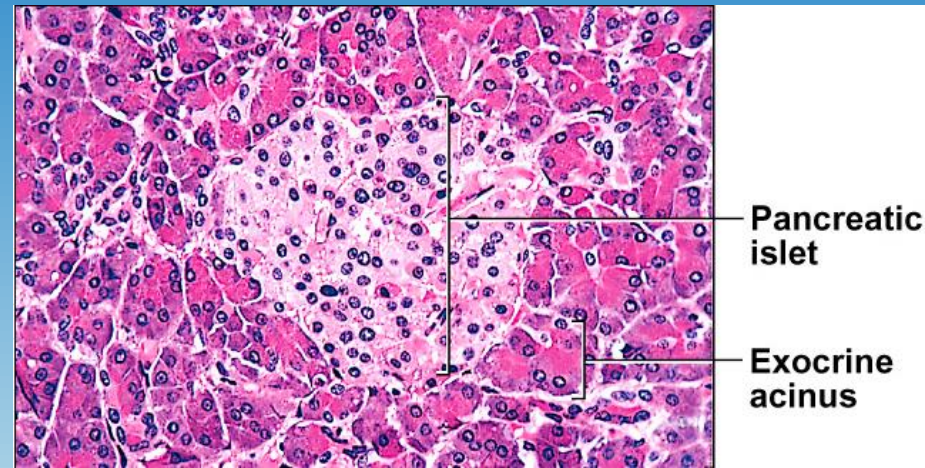
Pancreas



- Retroperitoneal, inferior and dorsal to stomach

Pancreatic Hormones

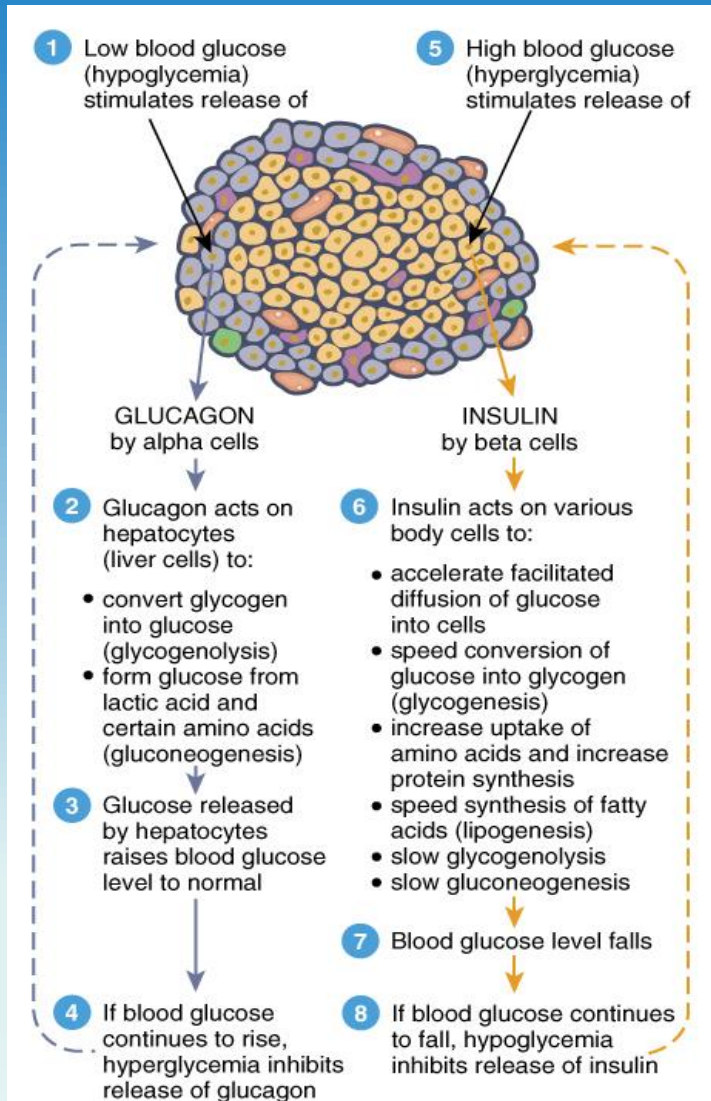
- 1-2 Million pancreatic islets producing hormones
 - 98% of organ produces
 - digestive enzymes
 - (exocrine)
- Insulin (from β cells)
 - secreted after meal with carbohydrates raises glucose blood levels
 - stimulates glucose and amino acid uptake
 - nutrient storage effect (stimulates glycogen, fat and protein synthesis)
 - antagonizes glucagon



Pancreatic Hormones 2

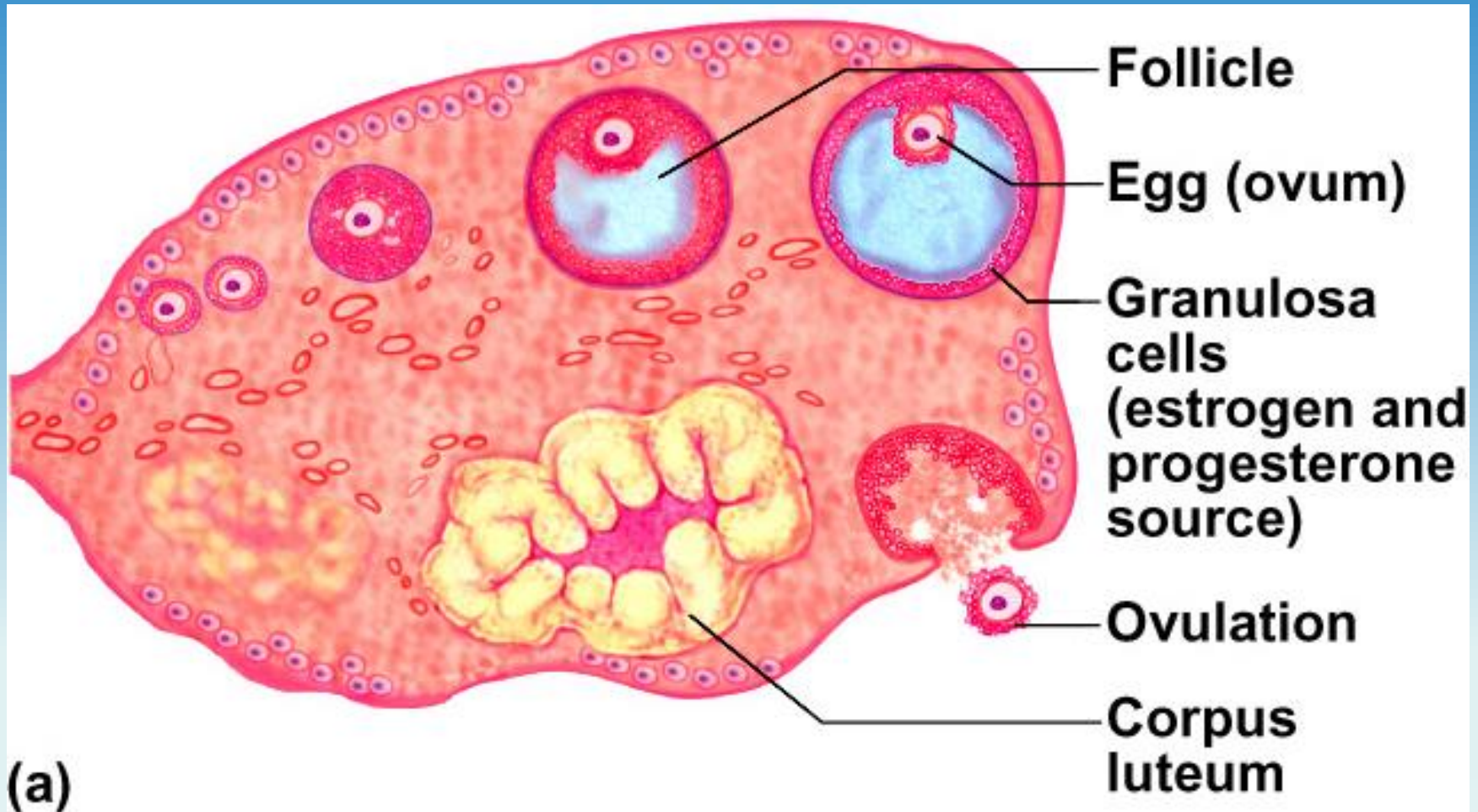
- Glucagon (from α cells)
 - secreted in very low carbohydrate and high protein diet or fasting
 - stimulates glycogenolysis, fat catabolism (release of FFA's) and promotes absorption of amino acids for gluconeogenesis
- Somatostatin from delta (δ) cells)
 - secreted with rise in blood glucose and amino acids after a meal
 - paracrine secretion = modulates secretion of $\alpha + \beta$ cells
- Hyperglycemic hormones raise blood glucose
 - glucagon, epinephrine, norepinephrine, cortisol & corticosterone
- Hypoglycemic hormones lower blood glucose
 - insulin

Regulation of Glucagon & Insulin Secretion

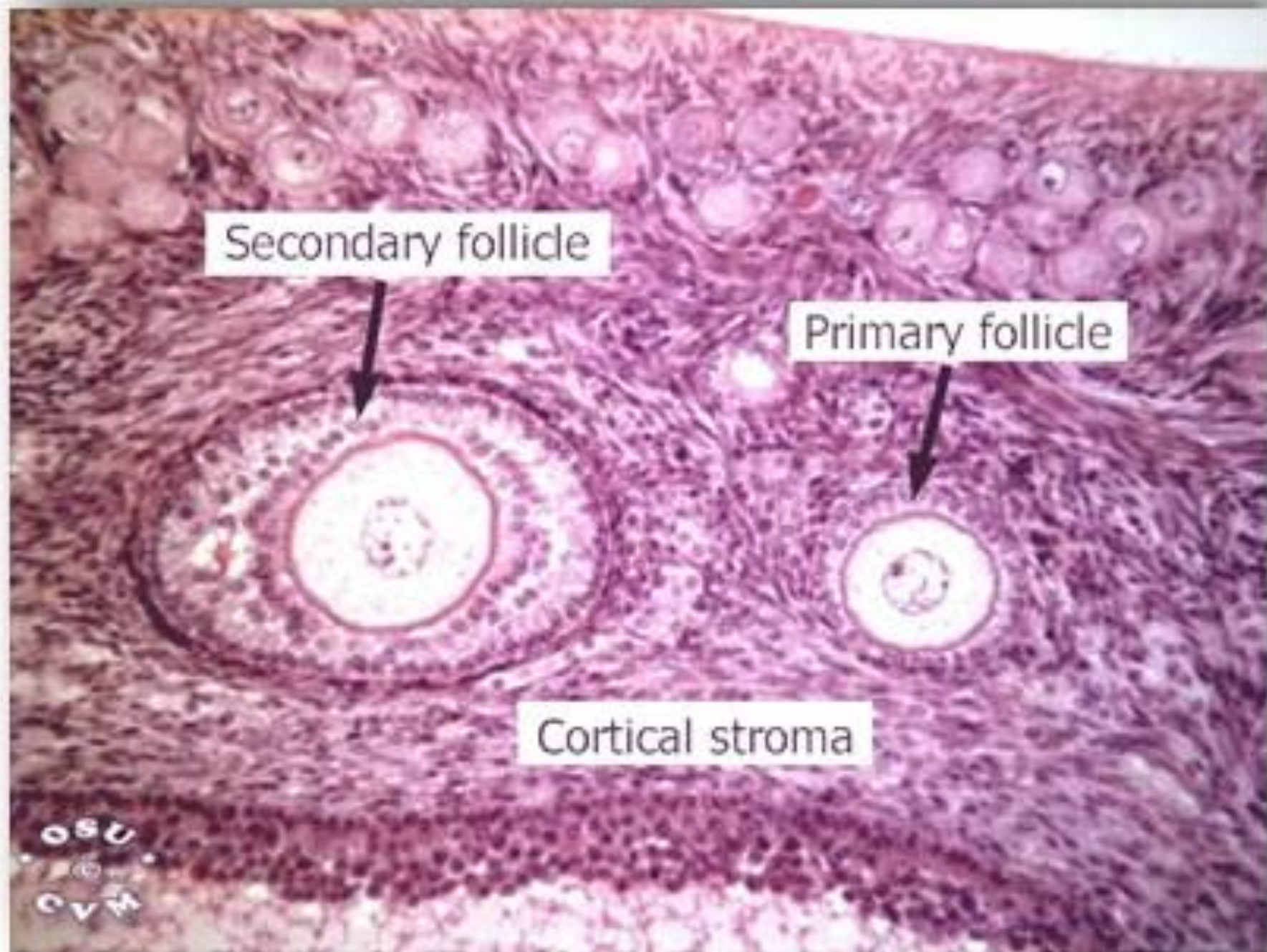


- Low blood glucose stimulates release of glucagon
- High blood glucose stimulates secretion of insulin

Histology of Ovary

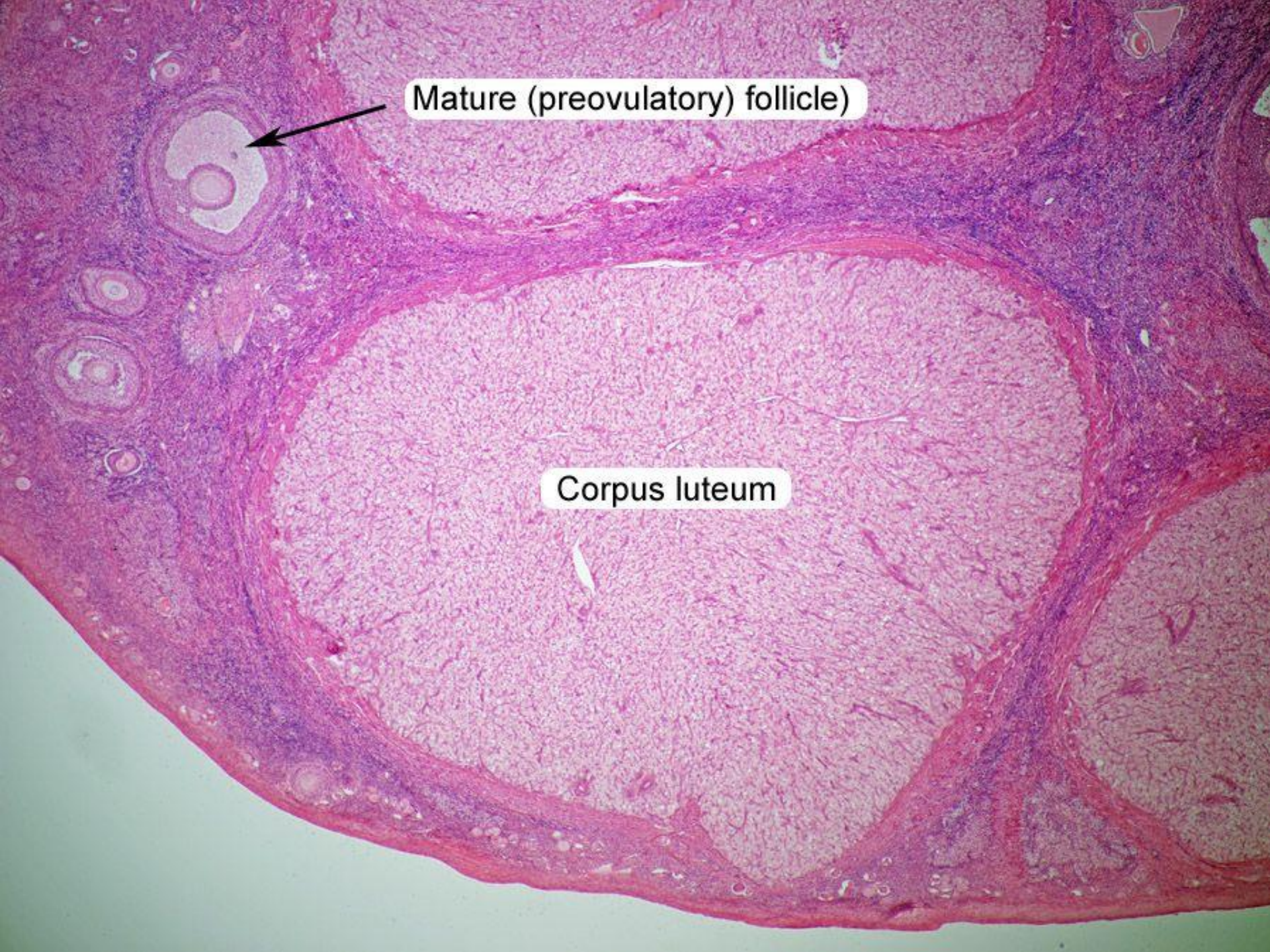


Follicles = egg surrounded by granulosa cells



Mature (preovulatory) follicle)

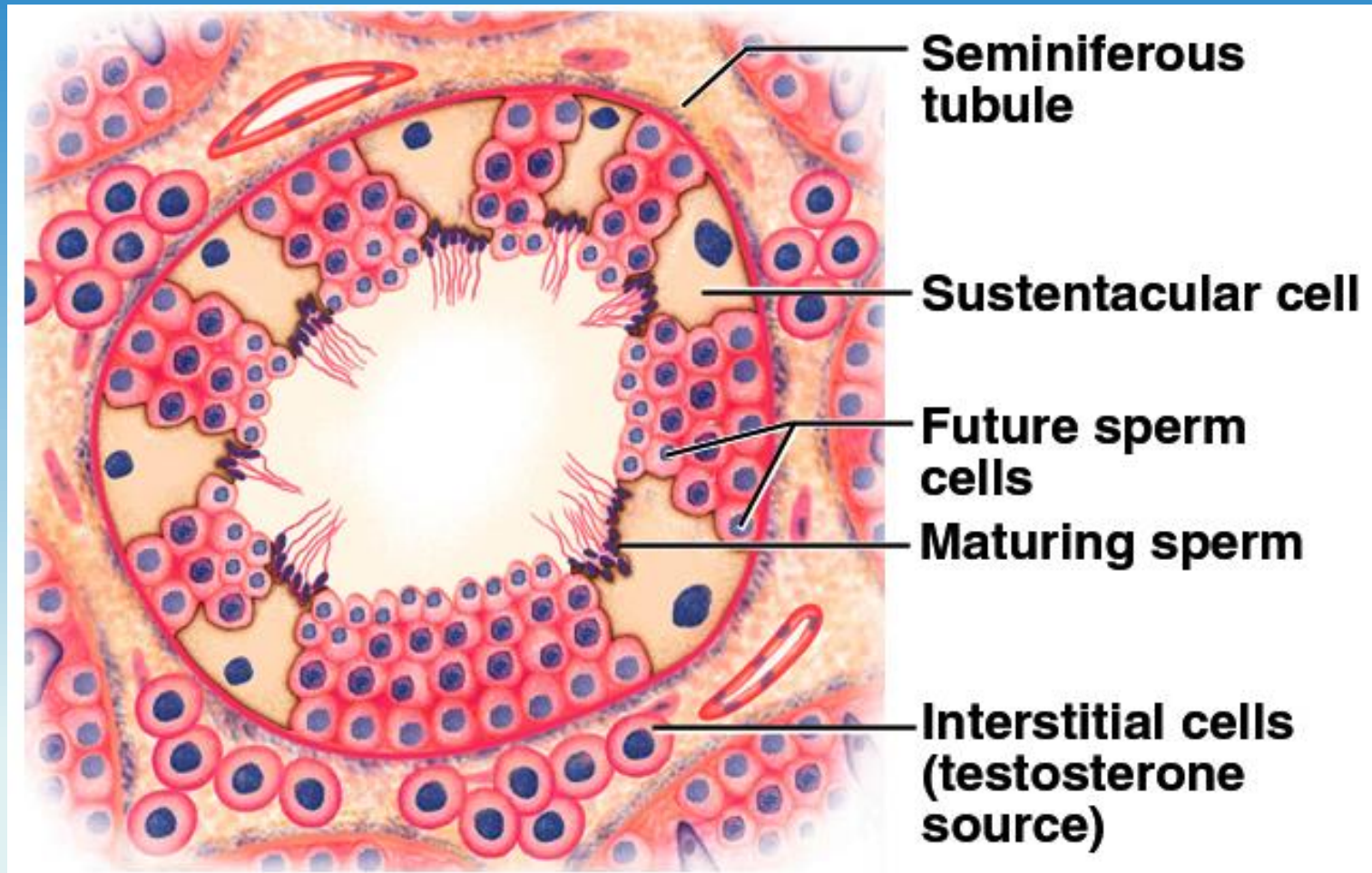
Corpus luteum



Ovary

- Granulosa cells in wall of ovarian follicle
 - produces estradiol, first half of menstrual cycle
- Corpus luteum: follicle after ovulation
 - produces estradiol and progesterone for 12 days or 8-12 weeks with pregnancy
- Functions of estradiol and progesterone
 - development of female reproductive system and physique including bone growth
 - regulate menstrual cycle, sustain pregnancy
 - prepare mammary glands for lactation
- Both secrete inhibin: suppresses FSH secretion

Histology of Testis



Seminiferous tubules produce sperm.

Testes

- Interstitial cells (between seminiferous tubules)
 - produce testosterone and estrogen
- Functions
 - development of male reproductive system and physique
 - sustains sperm production and sex drive
- Sustentacular sertoli cells
 - secrete inhibin which suppresses FSH secretion which stabilizes sperm production rates

Endocrine Functions of Other Organs

- Heart –
 - atrial natriuretic peptide released with an increase in BP
 - ↓ blood volume + ↓ BP by ↑ Na⁺ and H₂O loss by kidneys
- Skin
 - keratinocytes help produce D3, first step in synthesis
- Liver
 - converts vitamin D3 to calcidiol
 - source of IGF-I that works with GH
 - secretes about 15% of erythropoietin
 - secretes angiotensinogen (a prohormone)
 - precursor of angiotensin II, a vasoconstrictor

Endocrine Functions of Other Organs

- Kidneys
 - converts calcidiol to calcitriol (active form of vitamin D)
 - ↑ absorption by intestine and inhibits loss in the urine
 - more Ca^{+2} available for bone deposition
 - produces 85% of erythropoietin –
 - stimulates bone marrow to produce RBC's
 - convert angiotensinogen to angiotensin I
- Stomach and small intestines (10 enteric hormones)
 - coordinate digestive motility and secretion
- Placenta
 - secretes estrogen, progesterone and others
 - regulate pregnancy, stimulate development of fetus and mammary glands

Miscellaneous Hormones

Eicosanoids

- Local hormones released by all body cells
- Leukotrienes influence WBCs & inflammation
- Prostaglandins alter
 - smooth muscle contraction, glandular secretion, blood flow, platelet function, nerve transmission, metabolism etc.
- Ibuprofen & other nonsteroidal anti-inflammatory drugs treat pain, fever & inflammation by inhibiting prostaglandin synthesis

Nonsteroidal Anti-inflammatory Drugs

- Answer to how aspirin or ibuprofen works was discovered in 1971
 - inhibit a key enzyme in prostaglandin synthesis without affecting the synthesis of leukotrienes
- Treat a variety of inflammatory disorders
 - rheumatoid arthritis
- Usefulness of aspirin to treat fever & pain implies prostaglandins are responsible for those symptoms

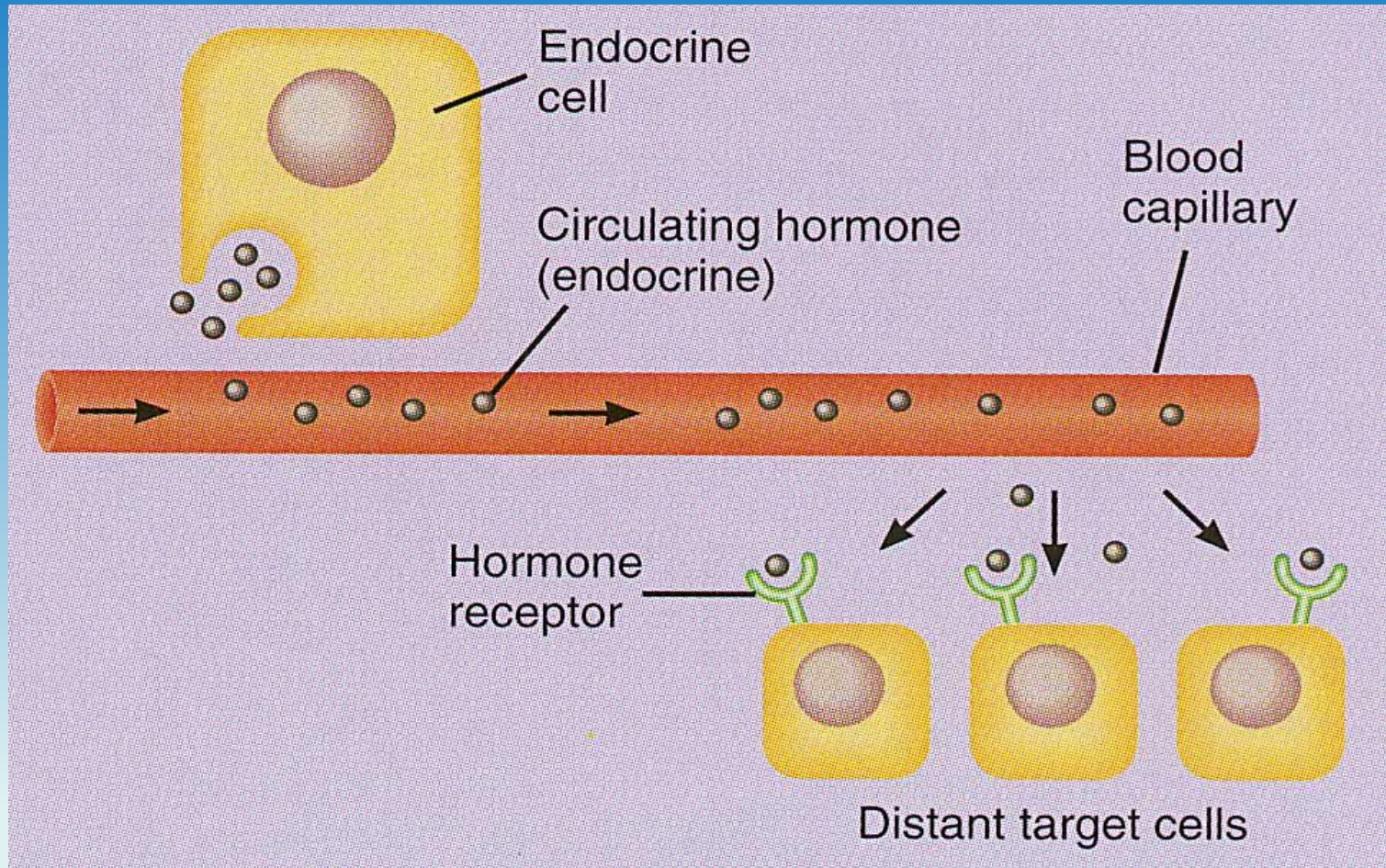
Growth Factors

- Substances with mitogenic qualities
 - cause cell growth from cell division
- Many act locally as autocrines or paracrines
- Selected list of growth factors (Table 18.12)
 - epidermal growth factor
 - platelet-derived growth factor
 - fibroblast growth factor
 - nerve growth factor
 - tumor angiogenesis factors
 - transforming growth factors

Hormone Transport

- Monoamines and peptides are hydrophilic so mix easily with blood plasma
- Steroids and thyroid hormone are hydrophobic and must bind to transport proteins for transport
 - **bound hormone** - hormone attached to transport protein, (prolongs half-life to weeks, protects from enzymes and kidney filtration)
 - only **unbound hormone** can leave capillary to reach target cell (half-life a few minutes)
- Transport proteins in blood plasma
 - albumin, thyretin and TGB (thyroxine binding globulin) bind to thyroid hormone
 - steroid hormones bind to globulins (transcortin)
 - aldosterone has no transport protein, 20 min. half-life

Hormone Receptors

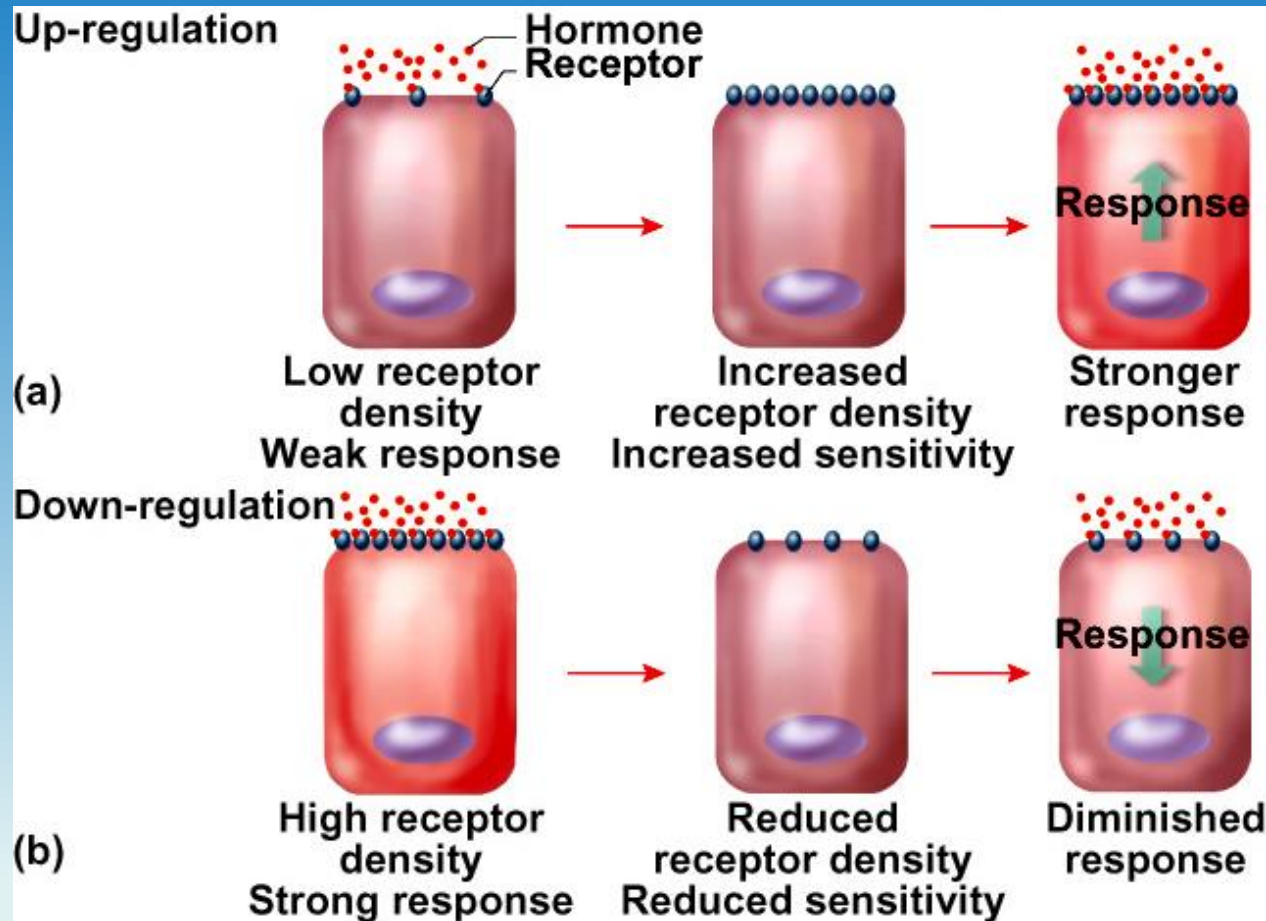


- Hormones only affect target cells with specific membrane proteins called receptors

Role of Hormone Receptors

- Constantly being synthesized & broken down
- A range of 2000-100,000 receptors / target cell
- Down-regulation
 - excess hormone, produces a decrease in number of receptors
 - receptors undergo endocytosis and are degraded
 - decreases sensitivity of target cell to hormone
- Up-regulation
 - deficiency of hormone, produces an increase in the number of receptors
 - target tissue more sensitive to the hormone

Modulation of Target Cell Sensitivity



- Long-term use of high pharmacological doses
 - bind to receptor sites of related hormones
 - target cell may convert to different hormone

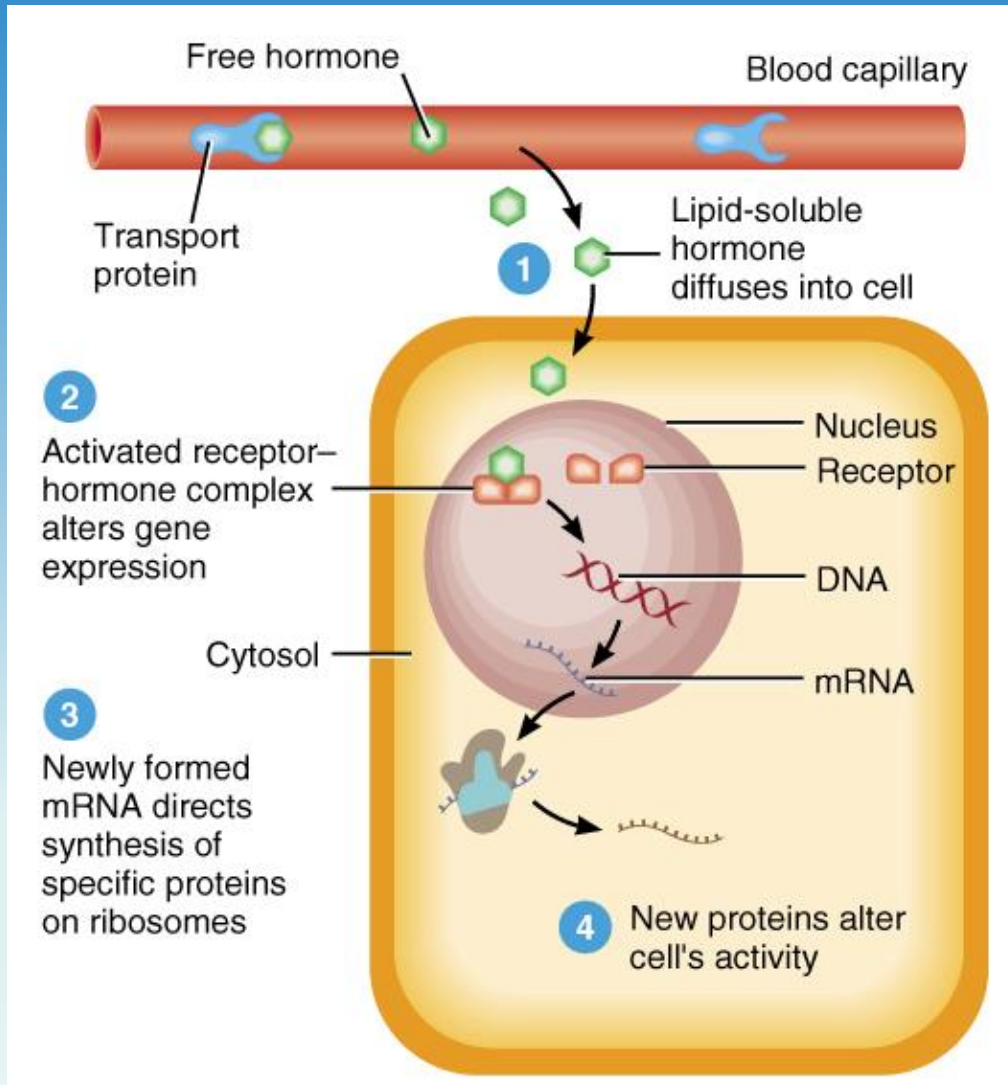
Blocking Hormone Receptors

- Synthetic hormones that block receptors for naturally occurring hormones
 - RU486 (mifepristone) binds to the receptors for progesterone preventing it from maintaining the uterus in a pregnant woman
 - used to induce abortion
 - brings on menstrual cycle
- Hormone is prevented from interacting with its receptors and can not perform its normal functions

General Mechanisms of Hormone Action

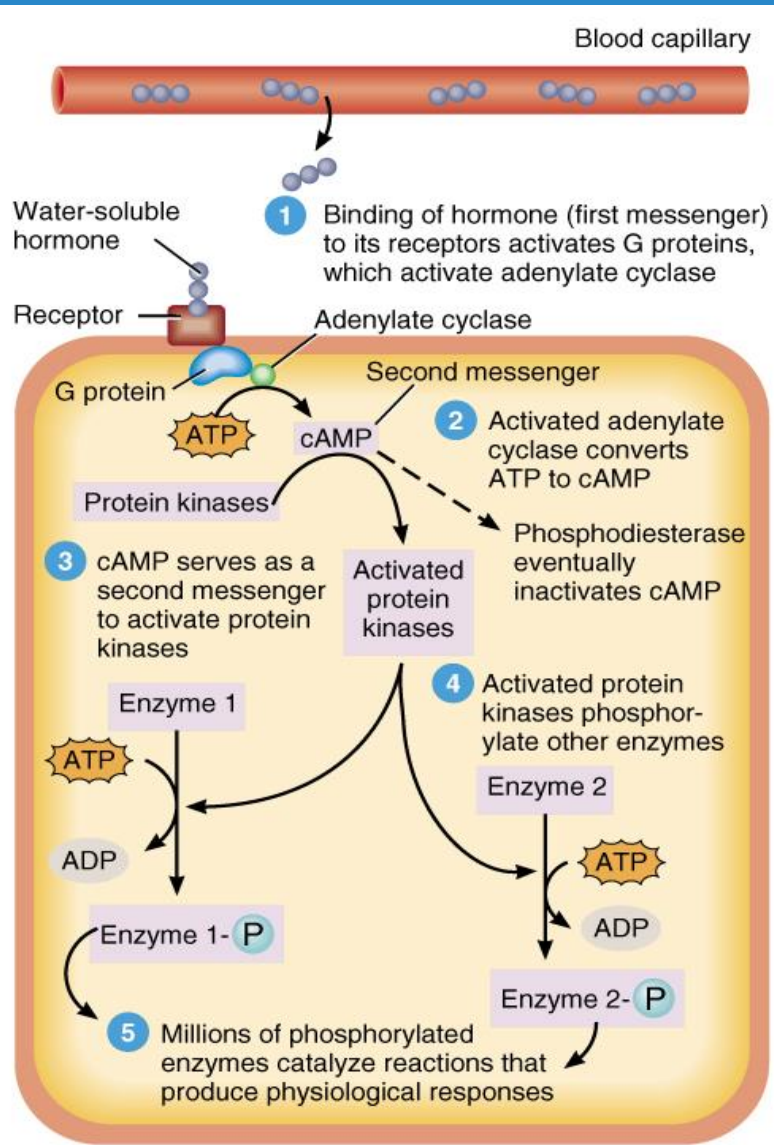
- Hormone binds to cell surface or receptor inside target cell
- Cell may then
 - synthesize new molecules
 - change permeability of membrane
 - alter rates of reactions
- Each target cell responds to hormone differently
 - liver cells---insulin stimulates glycogen synthesis
 - adipose---insulin stimulates triglyceride synthesis

Action of Lipid-Soluble Hormones



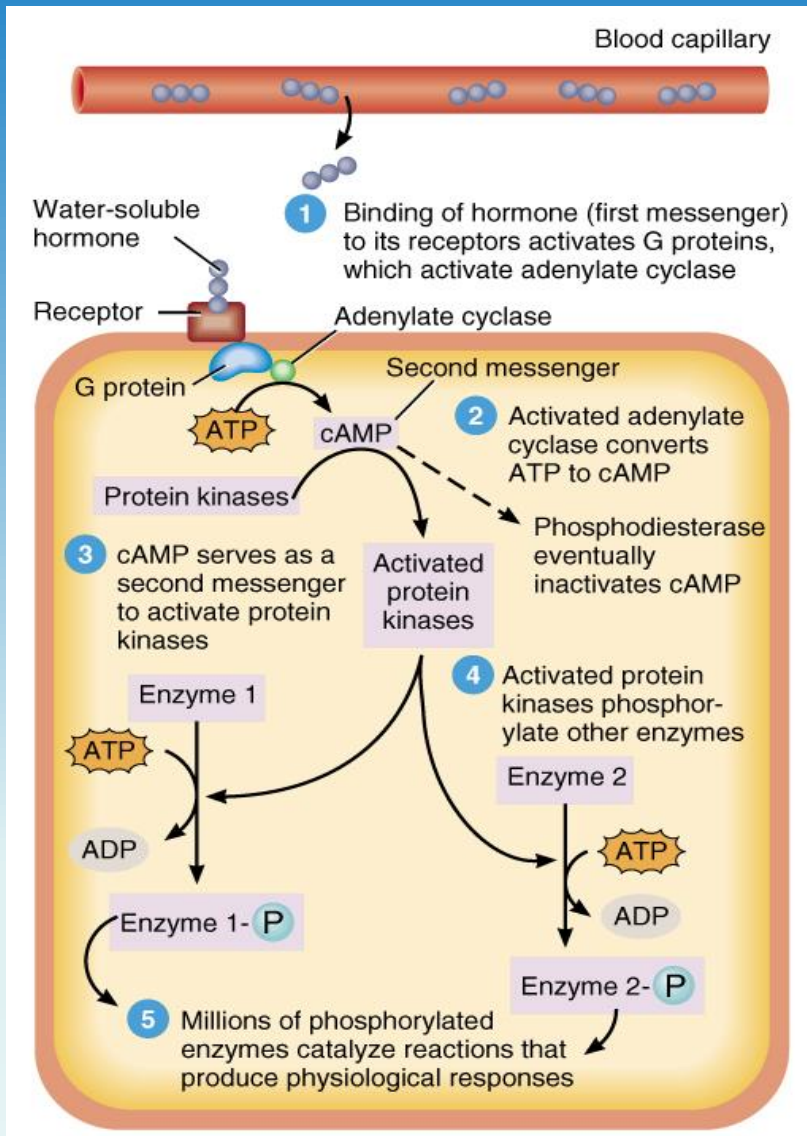
- Hormone diffuses through phospholipid bilayer & into cell
- Binds to receptor turning on/off specific genes
- New mRNA is formed & directs synthesis of new proteins
- New protein alters cell's activity

Action of Water-Soluble Hormones



- Can not diffuse through plasma membrane
- Hormone receptors are integral membrane proteins
 - act as first messenger
- Receptor protein activates G-protein in membrane
- G-protein activates adenylate cyclase to convert ATP to cAMP in the cytosol

Water-soluble Hormones (2)



- Cyclic AMP is the 2nd messenger
- Activates kinases in the cytosol to speed up/slow down physiological responses
- Phosphodiesterase inactivates cAMP quickly
- Cell response is turned off unless new hormone molecules arrive

Second Messengers

- Some hormones exert their influence by increasing the synthesis of cAMP
 - ADH, TSH, ACTH, glucagon and epinephrine
- Some exert their influence by decreasing the level of cAMP
 - growth hormone inhibiting hormone
- Other substances can act as 2nd messengers
 - calcium ions
 - cGMP
- Same hormone may use different 2nd messengers in different target cells

Amplification of Hormone Effects

- Single molecule of hormone binds to receptor
- Activates 100 G-proteins
- Each activates an adenylate cyclase molecule which then produces 1000 cAMP
- Each cAMP activates a protein kinase, which may act upon 1000's of substrate molecules
- One molecule of epinephrine may result in breakdown of millions of glycogen molecules into glucose molecules

Cholera Toxin and G Proteins

- Toxin is deadly because it produces massive watery diarrhea and person dies from dehydration
- Toxin of cholera bacteria causes G-protein to lock in activated state in intestinal epithelium
- Cyclic AMP causes intestinal cells to actively transport chloride (Na^+ and water follow) into the lumen
- Person die unless ions and fluids are replaced & receive antibiotic treatment

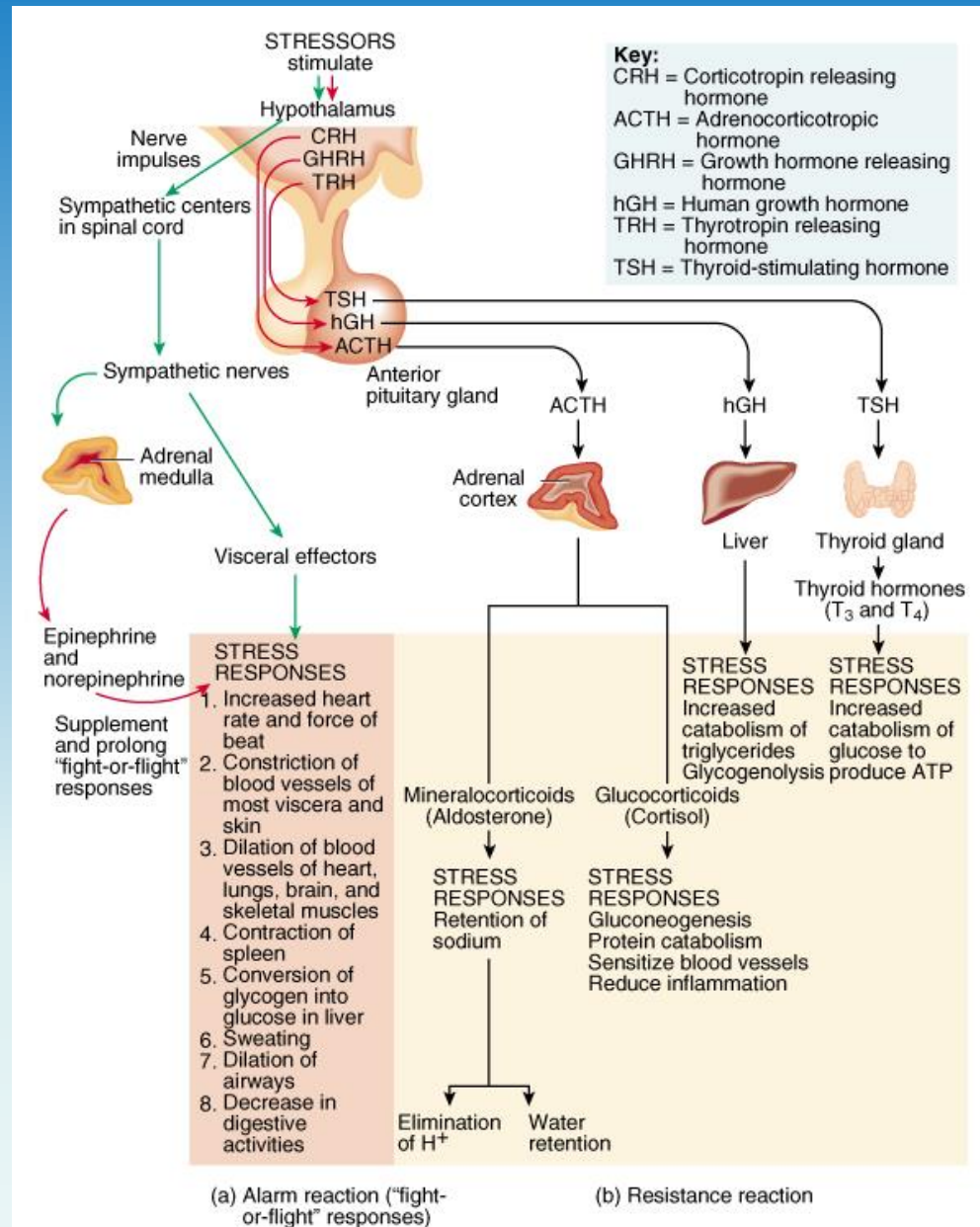
Hormonal Interactions

- Permissive effect
 - a second hormone, strengthens the effects of the first
 - thyroid strengthens epinephrine's effect upon lipolysis
- Synergistic effect
 - two hormones acting together for greater effect
 - estrogen & LH are both needed for oocyte production
- Antagonistic effects
 - two hormones with opposite effects
 - insulin promotes glycogen formation & glucagon stimulates glycogen breakdown

Stress & General Adaptation Syndrome

- Stress response is set of bodily changes called general adaptation syndrome (GAS)
- Any stimulus that produces a stress response is called a stressor
- Stress resets the body to meet an emergency
 - eustress is productive stress & helps us prepare for certain challenges
 - distress type levels of stress are harmful
 - lower our resistance to infection

General Adaptation Syndrome



Alarm Reaction (Fight-or-Flight)

- Initiated by hypothalamic stimulation of sympathetic portion of the ANS & adrenal medulla
- Dog attack
 - increases circulation
 - promotes ATP synthesis
 - nonessential body functions are inhibited
 - digestive, urinary & reproductive

Resistance Reaction

- Initiated by hypothalamic releasing hormones (long-term reaction to stress)
 - corticotropin, growth hormone & thyrotropin releasing hormones
- Results
 - increased secretion of aldosterone acts to conserve Na^+ (increases blood pressure) and eliminate H^+
 - increased secretion of cortisol so protein catabolism is increased & other sources of glucose are found
 - increase thyroid hormone to increase metabolism
- Allow body to continue to fight a stressor

Exhaustion

- Resources of the body have become depleted
- Resistance stage can not be maintained
- Prolonged exposure to resistance reaction hormones
 - wasting of muscle
 - suppression of immune system
 - ulceration of the GI tract
 - failure of the pancreatic beta cells

Stress and Disease

- Stress can lead to disease by inhibiting the immune system
 - hypertension, asthma, migraine, gastritis, colitis, and depression
- Interleukin - 1 is secreted by macrophages
 - link between stress and immunity
 - stimulates production of immune substances
 - feedback control since immune substance suppress the formation of interleukin-1

Aging and the Endocrine System

- Production of human growth hormone decreases
 - muscle atrophy
- Production of TSH increase with age to try and stimulate thyroid
 - decrease in metabolic rate, increase in body fat & hypothyroidism
- Thymus after puberty is replaced with adipose
- Adrenal glands produce less cortisol & aldosterone
- Receptor sensitivity to glucose declines
- Ovaries no longer respond to gonadotropins
 - decreased output of estrogen (osteoporosis & atherosclerosis)

Endocrine Disorders

- Variations in hormone concentration and target cell sensitivity have noticeable effects on the body
- Hyposecretion – inadequate hormone release
 - tumor or lesion destroys gland
 - head trauma affects pituitary gland's ability to secrete ADH
 - diabetes insipidus = chronic polyuria
- Hypersecretion – excessive hormone release
 - tumors or autoimmune disorder
 - toxic goiter (graves disease) – antibodies mimic effect of TSH on the thyroid

Pituitary Disorders

- Hypersecretion of growth hormones
 - acromegaly
 - thickening of the bones and soft tissues
 - problems in childhood or adolescence
 - gigantism if oversecretion
 - dwarfism if hyposecretion



Age 9



Age 16



Age 33



Age 52

Thyroid Gland Disorders

- Hyperthyroidism
 - Weight loss, tremors, nervousness, exophthalmos
- Endemic goiter (goiter = enlarged thyroid gland)
 - dietary iodine deficiency, no TH, no - feedback, $\uparrow$ TSH
- Toxic goiter (Graves disease)
 - antibodies mimic TSH, $\uparrow$ TH, exophthalmos



Thyroid Gland Disorders

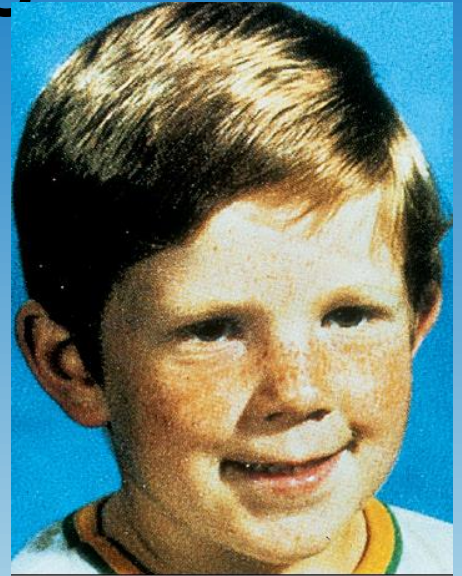
- Congenital hypothyroidism (↓ TH)
 - infant suffers abnormal bone development, thickened facial features, low temperature, lethargy, brain damage, dwarfism, cretinism
- Myxedema (adult hypothyroidism, ↓ **TH**)
 - low metabolic rate, sluggishness, sleepiness, weight gain, constipation, dry skin and hair, cold sensitivity, ↑ blood pressure and tissue swelling

Parathyroid Disorders

- Hypoparathyroid
 - surgical excision during thyroid surgery
 - fatal tetany 3-4 days
- Hyperparathyroid = excess PTH secretion
 - tumor in gland
 - causes soft, fragile and deformed bones
 - $\uparrow$ blood Ca^{+2}
 - renal calculi

Adrenal Disorders

- Cushing syndrome is excess cortical secretion
 - causes hyperglycemia, hypertension, weakness, edema
 - muscle and bone loss occurs with protein catabolism, poor wound healing
 - buffalo hump & moon face = fat deposition between shoulders or in face with spindly legs due to low fat
- Adrenogenital syndrome (AGS)
 - adrenal androgen hypersecretion accompanies Cushing
 - causes enlargement of external sexual organs in children & early onset of puberty
 - masculinizing effects on women (deeper voice & beard growth)



Addison's disease

- Hyposecretion of glucocorticoids
 - hypoglycemia, muscle weakness, low BP, dehydration due to decreased Na^+ in blood
 - mimics skin darkening effects of MSH
 - potential cardiac arrest

Diabetes Mellitus

- Signs and symptoms of hyposecretion of insulin
 - polyuria, polydipsia, polyphagia
 - hyperglycemia, glycosuria, ketonuria
 - osmotic diuresis : blood glucose levels rise above transport maximum of kidney tubules, glucose remains in urine, osmolarity ↑ and draws water into urine
- Transport maximum of glucose reabsorption
 - kidney tubules can not reabsorb glucose fast enough if no insulin is present
 - osmotic diuresis results due to excess glucose and ketones in tubules

Types of Diabetes Mellitus

- Type I (IDDM) - 10% of cases
 - some cases have autoimmune destruction of β cells, diagnosed about age 12
 - treated with diet, exercise, monitoring of blood glucose and periodic injections of insulin or insulin pump
- Type II (NIDDM) - 90%
 - insulin resistance
 - failure of target cells to respond to insulin
 - 3 major risk factors are heredity, age (40+) and obesity
 - treated with weight loss program of diet and exercise,
 - oral medications improve insulin secretion or target cell sensitivity

Pathology of Diabetes

- Acute pathology: cells cannot absorb glucose, rely on fat and proteins (weight loss + weakness)
 - fat catabolism $\uparrow$ FFA's in blood and ketone bodies
 - ketonuria promotes osmotic diuresis, loss of Na^+ + K^+
 - ketoacidosis occurs as ketones $\downarrow$ blood pH
 - if continued causes dyspnea and eventually diabetic coma
- Chronic pathology
 - chronic hyperglycemia leads to neuropathy and cardiovascular damage from atherosclerosis
 - retina and kidneys (common in type I), atherosclerosis leading to heart failure (common in type II), and gangrene

Hyperinsulinism

- From excess insulin injection or pancreatic islet tumor
- Causes hypoglycemia, weakness and hunger
 - triggers secretion of epinephrine, GH and glucagon
 - side effects: anxiety, sweating and $\uparrow$ HR
- Insulin shock
 - uncorrected hyperinsulinism with disorientation, convulsions or unconsciousness